SG No. 221 O_{h}^{1} $Pm\bar{3}m$ [cubic]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `1a`, site symmetry: `m-3m`

Table 1: Wyckoff bond: `3a@1a`, bond symmetry: `4/mm.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	[0, 0, 0]	`[1,-2,-3,4,17,-18,-19,20,-25,26,27,-28,-41,42,43,-44]`
2	$[0, X, 0]$	[0, 0, 0]	`[5,-6,-7,8,13,-14,-15,16,-29,30,31,-32,-37,38,39,-40]`
3	$[0, 0, X]$	[0, 0, 0]	`[9,-10,-11,12,-21,22,23,-24,-33,34,35,-36,45,-46,-47,48]`

Table 2: Wyckoff bond: `4b@1a`, bond symmetry: `.-3m`

No.	vector	center	mapping
1	$[X, X, X]$	[0, 0, 0]	`[1,5,9,-14,-19,-24,-25,-29,-33,38,43,48]`
2	$[-X, -X, X]$	[0, 0, 0]	`[2,7,12,-13,-17,-21,-26,-31,-36,37,41,45]`
3	$[-X, X, -X]$	[0, 0, 0]	`[3,8,10,-15,-20,-22,-27,-32,-34,39,44,46]`
4	$[X, -X, -X]$	[0, 0, 0]	`[4,6,11,-16,-18,-23,-28,-30,-35,40,42,47]`

Table 3: Wyckoff bond: `6c@1a`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	[0, 0, 0]	`[1,-2,13,-14,-25,26,-37,38]`
2	$[-X, X, 0]$	[0, 0, 0]	`[3,-4,-15,16,-27,28,39,-40]`
3	$[0, X, X]$	[0, 0, 0]	`[5,-6,23,-24,-29,30,-47,48]`
4	$[0, -X, X]$	[0, 0, 0]	`[7,-8,-21,22,-31,32,45,-46]`
5	$[X, 0, X]$	[0, 0, 0]	`[9,-10,-19,20,-33,34,43,-44]`
6	$[X, 0, -X]$	[0, 0, 0]	`[11,-12,17,-18,-35,36,-41,42]`

Table 4: Wyckoff bond: `12d@1a`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, Y]$	[0, 0, 0]	`[1,-14,-25,38]`
2	$[-X, -X, Y]$	[0, 0, 0]	`[2,-13,-26,37]`
3	$[-X, X, -Y]$	[0, 0, 0]	`[3,-15,-27,39]`
4	$[X, -X, -Y]$	[0, 0, 0]	`[4,-16,-28,40]`
5	$[Y, X, X]$	[0, 0, 0]	`[5,-24,-29,48]`
6	$[Y, -X, -X]$	[0, 0, 0]	`[6,-23,-30,47]`
7	$[-Y, -X, X]$	[0, 0, 0]	`[7,-21,-31,45]`
8	$[-Y, X, -X]$	[0, 0, 0]	`[8,-22,-32,46]`
9	$[X, Y, X]$	[0, 0, 0]	`[9,-19,-33,43]`
10	$[-X, Y, -X]$	[0, 0, 0]	`[10,-20,-34,44]`
11	$[X, -Y, -X]$	[0, 0, 0]	`[11,-18,-35,42]`
12	$[-X, -Y, X]$	[0, 0, 0]	`[12,-17,-36,41]`

Table 5: Wyckoff bond: `12e@1a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	[0, 0, 0]	`[1,-2,-25,26]`
2	$[-X, Y, 0]$	[0, 0, 0]	`[3,-4,-27,28]`
3	$[0, X, Y]$	[0, 0, 0]	`[5,-6,-29,30]`
4	$[0, -X, Y]$	[0, 0, 0]	`[7,-8,-31,32]`
5	$[Y, 0, X]$	[0, 0, 0]	`[9,-10,-33,34]`
6	$[Y, 0, -X]$	[0, 0, 0]	`[11,-12,-35,36]`
7	$[Y, X, 0]$	[0, 0, 0]	`[13,-14,-37,38]`
8	$[Y, -X, 0]$	[0, 0, 0]	`[15,-16,-39,40]`
9	$[X, 0, -Y]$	[0, 0, 0]	`[17,-18,-41,42]`
10	$[-X, 0, -Y]$	[0, 0, 0]	`[19,-20,-43,44]`
11	$[0, Y, -X]$	[0, 0, 0]	`[21,-22,-45,46]`
12	$[0, Y, X]$	[0, 0, 0]	`[23,-24,-47,48]`

Table 6: Wyckoff bond: `24f@1a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	[0, 0, 0]	`[1,-25]`
2	$[-X, -Y, Z]$	[0, 0, 0]	`[2,-26]`
3	$[-X, Y, -Z]$	[0, 0, 0]	`[3,-27]`
4	$[X, -Y, -Z]$	[0, 0, 0]	`[4,-28]`
5	$[Z, X, Y]$	[0, 0, 0]	`[5,-29]`
6	$[Z, -X, -Y]$	[0, 0, 0]	`[6,-30]`
7	$[-Z, -X, Y]$	[0, 0, 0]	`[7,-31]`
8	$[-Z, X, -Y]$	[0, 0, 0]	`[8,-32]`
9	$[Y, Z, X]$	[0, 0, 0]	`[9,-33]`
10	$[-Y, Z, -X]$	[0, 0, 0]	`[10,-34]`
11	$[Y, -Z, -X]$	[0, 0, 0]	`[11,-35]`
12	$[-Y, -Z, X]$	[0, 0, 0]	`[12,-36]`
13	$[Y, X, -Z]$	[0, 0, 0]	`[13,-37]`
14	$[-Y, -X, -Z]$	[0, 0, 0]	`[14,-38]`
15	$[Y, -X, Z]$	[0, 0, 0]	`[15,-39]`
16	$[-Y, X, Z]$	[0, 0, 0]	`[16,-40]`
17	$[X, Z, -Y]$	[0, 0, 0]	`[17,-41]`
18	$[-X, Z, Y]$	[0, 0, 0]	`[18,-42]`
19	$[-X, -Z, -Y]$	[0, 0, 0]	`[19,-43]`
20	$[X, -Z, Y]$	[0, 0, 0]	`[20,-44]`
21	$[Z, Y, -X]$	[0, 0, 0]	`[21,-45]`
22	$[Z, -Y, X]$	[0, 0, 0]	`[22,-46]`
23	$[-Z, Y, X]$	[0, 0, 0]	`[23,-47]`
24	$[-Z, -Y, -X]$	[0, 0, 0]	`[24,-48]`

* Wyckoff site: `1b`, site symmetry: `m-3m`

Table 7: Wyckoff bond: `3a@1b`, bond symmetry: `4/mm.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,-3,4,17,-18,-19,20,-25,26,27,-28,-41,42,43,-44]`
2	$[0, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-6,-7,8,13,-14,-15,16,-29,30,31,-32,-37,38,39,-40]`
3	$[0, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-10,-11,12,-21,22,23,-24,-33,34,35,-36,45,-46,-47,48]`

Table 8: Wyckoff bond: `4b@1b`, bond symmetry: `.-3m`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,5,9,-14,-19,-24,-25,-29,-33,38,43,48]`
2	$[-X, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2,7,12,-13,-17,-21,-26,-31,-36,37,41,45]`
3	$[-X, X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,8,10,-15,-20,-22,-27,-32,-34,39,44,46]`
4	$[X, -X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4,6,11,-16,-18,-23,-28,-30,-35,40,42,47]`

Table 9: Wyckoff bond: `6c@1b`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,13,-14,-25,26,-37,38]`
2	$[-X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-4,-15,16,-27,28,39,-40]`
3	$[0, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-6,23,-24,-29,30,-47,48]`
4	$[0, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[7,-8,-21,22,-31,32,45,-46]`
5	$[X, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-10,-19,20,-33,34,43,-44]`
6	$[X, 0, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[11,-12,17,-18,-35,36,-41,42]`

Table 10: Wyckoff bond: `12d@1b`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-14,-25,38]`
2	$[-X, -X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2,-13,-26,37]`
3	$[-X, X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-15,-27,39]`
4	$[X, -X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4,-16,-28,40]`
5	$[Y, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-24,-29,48]`
6	$[Y, -X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[6,-23,-30,47]`
7	$[-Y, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[7,-21,-31,45]`
8	$[-Y, X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[8,-22,-32,46]`
9	$[X, Y, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-19,-33,43]`
10	$[-X, Y, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[10,-20,-34,44]`
11	$[X, -Y, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[11,-18,-35,42]`
12	$[-X, -Y, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[12,-17,-36,41]`

Table 11: Wyckoff bond: `12e@1b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,-25,26]`

continued ...

Table 11

No.	vector	center	mapping
2	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,-27,28]`
3	$[0, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-6,-29,30]`
4	$[0, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-8,-31,32]`
5	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-10,-33,34]`
6	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[11,-12,-35,36]`
7	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,-14,-37,38]`
8	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[15,-16,-39,40]`
9	$[X, 0, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-18,-41,42]`
10	$[-X, 0, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-20,-43,44]`
11	$[0, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[21,-22,-45,46]`
12	$[0, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[23,-24,-47,48]`

Table 12: Wyckoff bond: `24f@1b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-27]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-28]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[8,-32]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[24,-48]`

* Wyckoff site: `3c`, site symmetry: `4/mm.m`

Table 13: Wyckoff bond: `3a@3c`, bond symmetry: `4/mm.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,-3,4,17,-18,-19,20,-25,26,27,-28,-41,42,43,-44]`
2	$[0, X, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,-6,-7,8,13,-14,-15,16,-29,30,31,-32,-37,38,39,-40]`
3	$[0, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9,-10,-11,12,-21,22,23,-24,-33,34,35,-36,45,-46,-47,48]`

Table 14: Wyckoff bond: `6b@3c`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[0, X, X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[1,-4,18,-19,-25,28,-42,43]`
2	$[0, -X, X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,-3,-17,20,-26,27,41,-44]`
3	$[X, 0, X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,-8,-14,15,-29,32,38,-39]`
4	$[X, 0, -X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[6,-7,13,-16,-30,31,-37,40]`
5	$[X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9,-12,21,-24,-33,36,-45,48]`
6	$[-X, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[10,-11,-22,23,-34,35,46,-47]`

Table 15: Wyckoff bond: `6c@3c`, bond symmetry: `mmm..`

No.	vector	center	mapping
1	$[0, 0, X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[1,2,-3,-4,-25,-26,27,28]`
2	$[X, 0, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,6,-7,-8,-29,-30,31,32]`
3	$[0, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9,10,-11,-12,-33,-34,35,36]`
4	$[0, 0, -X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[13,14,-15,-16,-37,-38,39,40]`
5	$[0, X, 0]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[17,18,-19,-20,-41,-42,43,44]`
6	$[X, 0, 0]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[21,22,-23,-24,-45,-46,47,48]`

Table 16: Wyckoff bond: `12d@3c`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[1,-19,-25,43]`
2	$[-Y, -X, X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[2,-17,-26,41]`
3	$[-Y, X, -X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[3,-20,-27,44]`
4	$[Y, -X, -X]$	$[0, \frac{1}{2}, \frac{1}{2}]$	`[4,-18,-28,42]`
5	$[X, Y, X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,-14,-29,38]`
6	$[X, -Y, -X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[6,-16,-30,40]`
7	$[-X, -Y, X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[7,-13,-31,37]`
8	$[-X, Y, -X]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[8,-15,-32,39]`
9	$[X, X, Y]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[9,-24,-33,48]`
10	$[-X, X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[10,-22,-34,46]`
11	$[X, -X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[11,-23,-35,47]`
12	$[-X, -X, Y]$	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[12,-21,-36,45]`

Table 17: Wyckoff bond: `12e@3c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-25,26]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,-27,28]`
3	$[0, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,-29,30]`
4	$[0, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-8,-31,32]`
5	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,-33,34]`
6	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-12,-35,36]`
7	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,-14,-37,38]`
8	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[15,-16,-39,40]`
9	$[X, 0, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-18,-41,42]`
10	$[-X, 0, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-20,-43,44]`
11	$[0, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,-22,-45,46]`
12	$[0, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[23,-24,-47,48]`

Table 18: Wyckoff bond: `12f@3c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,-25,28]`
2	$[0, -X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-26,27]`
3	$[Y, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-8,-29,32]`
4	$[Y, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-7,-30,31]`
5	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-12,-33,36]`
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-11,-34,35]`
7	$[X, 0, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,-16,-37,40]`
8	$[-X, 0, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[14,-15,-38,39]`
9	$[0, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-20,-41,44]`
10	$[0, Y, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[18,-19,-42,43]`
11	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,-24,-45,48]`
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[22,-23,-46,47]`

Table 19: Wyckoff bond: `24g@3c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-27]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-28]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-32]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-36]`

continued ...

Table 19

No.	vector	center	mapping
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[24,-48]`

* Wyckoff site: `3d`, site symmetry: `4/mm.m`

Table 20: Wyckoff bond: `3a@3d`, bond symmetry: `4/mm.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-2,-3,4,17,-18,-19,20,-25,26,27,-28,-41,42,43,-44]`
2	$[0, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-6,-7,8,13,-14,-15,16,-29,30,31,-32,-37,38,39,-40]`
3	$[0, 0, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[9,-10,-11,12,-21,22,23,-24,-33,34,35,-36,45,-46,-47,48]`

Table 21: Wyckoff bond: `6b@3d`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,-4,18,-19,-25,28,-42,43]`
2	$[0, -X, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-3,-17,20,-26,27,41,-44]`
3	$[X, 0, X]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,-8,-14,15,-29,32,38,-39]`
4	$[X, 0, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[6,-7,13,-16,-30,31,-37,40]`
5	$[X, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[9,-12,21,-24,-33,36,-45,48]`
6	$[-X, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[10,-11,-22,23,-34,35,46,-47]`

Table 22: Wyckoff bond: `6c@3d`, bond symmetry: `mmm..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,-3,-4,-25,-26,27,28]`
2	$[X, 0, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[5,6,-7,-8,-29,-30,31,32]`
3	$[0, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,-11,-12,-33,-34,35,36]`
4	$[0, 0, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,14,-15,-16,-37,-38,39,40]`
5	$[0, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,18,-19,-20,-41,-42,43,44]`
6	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,22,-23,-24,-45,-46,47,48]`

Table 23: Wyckoff bond: `12d@3d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[\frac{1}{2}, 0, 0]$	`[1,-19,-25,43]`
2	$[-Y, -X, X]$	$[\frac{1}{2}, 0, 0]$	`[2,-17,-26,41]`
3	$[-Y, X, -X]$	$[\frac{1}{2}, 0, 0]$	`[3,-20,-27,44]`
4	$[Y, -X, -X]$	$[\frac{1}{2}, 0, 0]$	`[4,-18,-28,42]`
5	$[X, Y, X]$	$[0, \frac{1}{2}, 0]$	`[5,-14,-29,38]`
6	$[X, -Y, -X]$	$[0, \frac{1}{2}, 0]$	`[6,-16,-30,40]`
7	$[-X, -Y, X]$	$[0, \frac{1}{2}, 0]$	`[7,-13,-31,37]`
8	$[-X, Y, -X]$	$[0, \frac{1}{2}, 0]$	`[8,-15,-32,39]`
9	$[X, X, Y]$	$[0, 0, \frac{1}{2}]$	`[9,-24,-33,48]`
10	$[-X, X, -Y]$	$[0, 0, \frac{1}{2}]$	`[10,-22,-34,46]`
11	$[X, -X, -Y]$	$[0, 0, \frac{1}{2}]$	`[11,-23,-35,47]`
12	$[-X, -X, Y]$	$[0, 0, \frac{1}{2}]$	`[12,-21,-36,45]`

Table 24: Wyckoff bond: `12e@3d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, 0, 0]$	`[1,-2,-25,26]`
2	$[-X, Y, 0]$	$[\frac{1}{2}, 0, 0]$	`[3,-4,-27,28]`
3	$[0, X, Y]$	$[0, \frac{1}{2}, 0]$	`[5,-6,-29,30]`
4	$[0, -X, Y]$	$[0, \frac{1}{2}, 0]$	`[7,-8,-31,32]`
5	$[Y, 0, X]$	$[0, 0, \frac{1}{2}]$	`[9,-10,-33,34]`
6	$[Y, 0, -X]$	$[0, 0, \frac{1}{2}]$	`[11,-12,-35,36]`
7	$[Y, X, 0]$	$[0, \frac{1}{2}, 0]$	`[13,-14,-37,38]`
8	$[Y, -X, 0]$	$[0, \frac{1}{2}, 0]$	`[15,-16,-39,40]`
9	$[X, 0, -Y]$	$[\frac{1}{2}, 0, 0]$	`[17,-18,-41,42]`
10	$[-X, 0, -Y]$	$[\frac{1}{2}, 0, 0]$	`[19,-20,-43,44]`
11	$[0, Y, -X]$	$[0, 0, \frac{1}{2}]$	`[21,-22,-45,46]`
12	$[0, Y, X]$	$[0, 0, \frac{1}{2}]$	`[23,-24,-47,48]`

Table 25: Wyckoff bond: `12f@3d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[\frac{1}{2}, 0, 0]$	`[1,-4,-25,28]`
2	$[0, -X, Y]$	$[\frac{1}{2}, 0, 0]$	`[2,-3,-26,27]`
3	$[Y, 0, X]$	$[0, \frac{1}{2}, 0]$	`[5,-8,-29,32]`
4	$[Y, 0, -X]$	$[0, \frac{1}{2}, 0]$	`[6,-7,-30,31]`
5	$[X, Y, 0]$	$[0, 0, \frac{1}{2}]$	`[9,-12,-33,36]`
6	$[-X, Y, 0]$	$[0, 0, \frac{1}{2}]$	`[10,-11,-34,35]`
7	$[X, 0, -Y]$	$[0, \frac{1}{2}, 0]$	`[13,-16,-37,40]`
8	$[-X, 0, -Y]$	$[0, \frac{1}{2}, 0]$	`[14,-15,-38,39]`
9	$[0, Y, -X]$	$[\frac{1}{2}, 0, 0]$	`[17,-20,-41,44]`
10	$[0, Y, X]$	$[\frac{1}{2}, 0, 0]$	`[18,-19,-42,43]`
11	$[Y, X, 0]$	$[0, 0, \frac{1}{2}]$	`[21,-24,-45,48]`
12	$[Y, -X, 0]$	$[0, 0, \frac{1}{2}]$	`[22,-23,-46,47]`

Table 26: Wyckoff bond: `24g@3d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[1,-25]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[2,-26]`
3	$[-X, Y, -Z]$	$[\frac{1}{2}, 0, 0]$	`[3,-27]`
4	$[X, -Y, -Z]$	$[\frac{1}{2}, 0, 0]$	`[4,-28]`
5	$[Z, X, Y]$	$[0, \frac{1}{2}, 0]$	`[5,-29]`
6	$[Z, -X, -Y]$	$[0, \frac{1}{2}, 0]$	`[6,-30]`
7	$[-Z, -X, Y]$	$[0, \frac{1}{2}, 0]$	`[7,-31]`
8	$[-Z, X, -Y]$	$[0, \frac{1}{2}, 0]$	`[8,-32]`
9	$[Y, Z, X]$	$[0, 0, \frac{1}{2}]$	`[9,-33]`
10	$[-Y, Z, -X]$	$[0, 0, \frac{1}{2}]$	`[10,-34]`
11	$[Y, -Z, -X]$	$[0, 0, \frac{1}{2}]$	`[11,-35]`
12	$[-Y, -Z, X]$	$[0, 0, \frac{1}{2}]$	`[12,-36]`
13	$[Y, X, -Z]$	$[0, \frac{1}{2}, 0]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$[0, \frac{1}{2}, 0]$	`[14,-38]`
15	$[Y, -X, Z]$	$[0, \frac{1}{2}, 0]$	`[15,-39]`
16	$[-Y, X, Z]$	$[0, \frac{1}{2}, 0]$	`[16,-40]`
17	$[X, Z, -Y]$	$[\frac{1}{2}, 0, 0]$	`[17,-41]`
18	$[-X, Z, Y]$	$[\frac{1}{2}, 0, 0]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$[\frac{1}{2}, 0, 0]$	`[19,-43]`
20	$[X, -Z, Y]$	$[\frac{1}{2}, 0, 0]$	`[20,-44]`
21	$[Z, Y, -X]$	$[0, 0, \frac{1}{2}]$	`[21,-45]`
22	$[Z, -Y, X]$	$[0, 0, \frac{1}{2}]$	`[22,-46]`
23	$[-Z, Y, X]$	$[0, 0, \frac{1}{2}]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$[0, 0, \frac{1}{2}]$	`[24,-48]`

* Wyckoff site: `6e`, site symmetry: `4m.m`

Table 27: Wyckoff bond: `6a@6e`, bond symmetry: `4m.m`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4,17,20,26,27,42,43]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,3,18,19,25,28,41,44]`
3	$[0, X, 0]$	$[0, x, 0]$	`[5,8,13,16,30,31,38,39]`
4	$[0, -X, 0]$	$[0, -x, 0]$	`[6,7,14,15,29,32,37,40]`
5	$[0, 0, X]$	$[0, 0, x]$	`[9,12,22,23,34,35,45,48]`
6	$[0, 0, -X]$	$[0, 0, -x]$	`[10,11,21,24,33,36,46,47]`

Table 28: Wyckoff bond: `12b@6e`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, X, X]$	$[x, 0, 0]$	`[1,-4,-42,43]`
2	$[0, -X, X]$	$[-x, 0, 0]$	`[2,-3,41,-44]`
3	$[X, 0, X]$	$[0, x, 0]$	`[5,-8,38,-39]`
4	$[X, 0, -X]$	$[0, -x, 0]$	`[6,-7,-37,40]`
5	$[X, X, 0]$	$[0, 0, x]$	`[9,-12,-45,48]`

continued ...

Table 28

No.	vector	center	mapping
6	$[-X, X, 0]$	$[0, 0, -x]$	`[10,-11,46,-47]`
7	$[X, 0, -X]$	$[0, x, 0]$	`[13,-16,-30,31]`
8	$[-X, 0, -X]$	$[0, -x, 0]$	`[14,-15,29,-32]`
9	$[0, X, -X]$	$[x, 0, 0]$	`[17,-20,26,-27]`
10	$[0, X, X]$	$[-x, 0, 0]$	`[18,-19,-25,28]`
11	$[X, X, 0]$	$[0, 0, -x]$	`[21,-24,-33,36]`
12	$[X, -X, 0]$	$[0, 0, x]$	`[22,-23,34,-35]`

Table 29: Wyckoff bond: `12c@6e`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, 0, X]$	$[x, 0, 0]$	`[1,-4,-26,27]`
2	$[0, 0, X]$	$[-x, 0, 0]$	`[2,-3,-25,28]`
3	$[X, 0, 0]$	$[0, x, 0]$	`[5,-8,-30,31]`
4	$[X, 0, 0]$	$[0, -x, 0]$	`[6,-7,-29,32]`
5	$[0, X, 0]$	$[0, 0, x]$	`[9,-12,-34,35]`
6	$[0, X, 0]$	$[0, 0, -x]$	`[10,-11,-33,36]`
7	$[0, 0, -X]$	$[0, x, 0]$	`[13,-16,-38,39]`
8	$[0, 0, -X]$	$[0, -x, 0]$	`[14,-15,-37,40]`
9	$[0, X, 0]$	$[x, 0, 0]$	`[17,-20,-42,43]`
10	$[0, X, 0]$	$[-x, 0, 0]$	`[18,-19,-41,44]`
11	$[X, 0, 0]$	$[0, 0, -x]$	`[21,-24,-46,47]`
12	$[X, 0, 0]$	$[0, 0, x]$	`[22,-23,-45,48]`

Table 30: Wyckoff bond: `24d@6e`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[x, 0, 0]$	`[1,43]`
2	$[-Y, -X, X]$	$[-x, 0, 0]$	`[2,41]`
3	$[-Y, X, -X]$	$[-x, 0, 0]$	`[3,44]`
4	$[Y, -X, -X]$	$[x, 0, 0]$	`[4,42]`
5	$[X, Y, X]$	$[0, x, 0]$	`[5,38]`
6	$[X, -Y, -X]$	$[0, -x, 0]$	`[6,40]`
7	$[-X, -Y, X]$	$[0, -x, 0]$	`[7,37]`
8	$[-X, Y, -X]$	$[0, x, 0]$	`[8,39]`
9	$[X, X, Y]$	$[0, 0, x]$	`[9,48]`
10	$[-X, X, -Y]$	$[0, 0, -x]$	`[10,46]`
11	$[X, -X, -Y]$	$[0, 0, -x]$	`[11,47]`
12	$[-X, -X, Y]$	$[0, 0, x]$	`[12,45]`
13	$[X, Y, -X]$	$[0, x, 0]$	`[13,31]`
14	$[-X, -Y, -X]$	$[0, -x, 0]$	`[14,29]`
15	$[X, -Y, X]$	$[0, -x, 0]$	`[15,32]`
16	$[-X, Y, X]$	$[0, x, 0]$	`[16,30]`
17	$[Y, X, -X]$	$[x, 0, 0]$	`[17,26]`

continued ...

Table 30

No.	vector	center	mapping
18	$[-Y, X, X]$	$[-x, 0, 0]$	`[18,28]`
19	$[-Y, -X, -X]$	$[-x, 0, 0]$	`[19,25]`
20	$[Y, -X, X]$	$[x, 0, 0]$	`[20,27]`
21	$[X, X, -Y]$	$[0, 0, -x]$	`[21,36]`
22	$[X, -X, Y]$	$[0, 0, x]$	`[22,34]`
23	$[-X, X, Y]$	$[0, 0, x]$	`[23,35]`
24	$[-X, -X, -Y]$	$[0, 0, -x]$	`[24,33]`

Table 31: Wyckoff bond: `24e@6e`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, 0, 0]$	`[1,26]`
2	$[-X, -Y, 0]$	$[-x, 0, 0]$	`[2,25]`
3	$[-X, Y, 0]$	$[-x, 0, 0]$	`[3,28]`
4	$[X, -Y, 0]$	$[x, 0, 0]$	`[4,27]`
5	$[0, X, Y]$	$[0, x, 0]$	`[5,30]`
6	$[0, -X, -Y]$	$[0, -x, 0]$	`[6,29]`
7	$[0, -X, Y]$	$[0, -x, 0]$	`[7,32]`
8	$[0, X, -Y]$	$[0, x, 0]$	`[8,31]`
9	$[Y, 0, X]$	$[0, 0, x]$	`[9,34]`
10	$[-Y, 0, -X]$	$[0, 0, -x]$	`[10,33]`
11	$[Y, 0, -X]$	$[0, 0, -x]$	`[11,36]`
12	$[-Y, 0, X]$	$[0, 0, x]$	`[12,35]`
13	$[Y, X, 0]$	$[0, x, 0]$	`[13,38]`
14	$[-Y, -X, 0]$	$[0, -x, 0]$	`[14,37]`
15	$[Y, -X, 0]$	$[0, -x, 0]$	`[15,40]`
16	$[-Y, X, 0]$	$[0, x, 0]$	`[16,39]`
17	$[X, 0, -Y]$	$[x, 0, 0]$	`[17,42]`
18	$[-X, 0, Y]$	$[-x, 0, 0]$	`[18,41]`
19	$[-X, 0, -Y]$	$[-x, 0, 0]$	`[19,44]`
20	$[X, 0, Y]$	$[x, 0, 0]$	`[20,43]`
21	$[0, Y, -X]$	$[0, 0, -x]$	`[21,46]`
22	$[0, -Y, X]$	$[0, 0, x]$	`[22,45]`
23	$[0, Y, X]$	$[0, 0, x]$	`[23,48]`
24	$[0, -Y, -X]$	$[0, 0, -x]$	`[24,47]`

Table 32: Wyckoff bond: `24f@6e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$[-x, 0, 0]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, x, 0]$	`[5,-8]`
4	$[Y, 0, -X]$	$[0, -x, 0]$	`[6,-7]`
5	$[X, Y, 0]$	$[0, 0, x]$	`[9,-12]`

continued ...

Table 32

No.	vector	center	mapping
6	$[-X, Y, 0]$	$[0, 0, -x]$	[10,-11]
7	$[X, 0, -Y]$	$[0, x, 0]$	[13,-16]
8	$[-X, 0, -Y]$	$[0, -x, 0]$	[14,-15]
9	$[0, Y, -X]$	$[x, 0, 0]$	[17,-20]
10	$[0, Y, X]$	$[-x, 0, 0]$	[18,-19]
11	$[Y, X, 0]$	$[0, 0, -x]$	[21,-24]
12	$[Y, -X, 0]$	$[0, 0, x]$	[22,-23]
13	$[0, -X, -Y]$	$[-x, 0, 0]$	[25,-28]
14	$[0, X, -Y]$	$[x, 0, 0]$	[26,-27]
15	$[-Y, 0, -X]$	$[0, -x, 0]$	[29,-32]
16	$[-Y, 0, X]$	$[0, x, 0]$	[30,-31]
17	$[-X, -Y, 0]$	$[0, 0, -x]$	[33,-36]
18	$[X, -Y, 0]$	$[0, 0, x]$	[34,-35]
19	$[-X, 0, Y]$	$[0, -x, 0]$	[37,-40]
20	$[X, 0, Y]$	$[0, x, 0]$	[38,-39]
21	$[0, -Y, X]$	$[-x, 0, 0]$	[41,-44]
22	$[0, -Y, -X]$	$[x, 0, 0]$	[42,-43]
23	$[-Y, -X, 0]$	$[0, 0, x]$	[45,-48]
24	$[-Y, X, 0]$	$[0, 0, -x]$	[46,-47]

Table 33: Wyckoff bond: 48g@6e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	[1]
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	[2]
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	[3]
4	$[X, -Y, -Z]$	$[x, 0, 0]$	[4]
5	$[Z, X, Y]$	$[0, x, 0]$	[5]
6	$[Z, -X, -Y]$	$[0, -x, 0]$	[6]
7	$[-Z, -X, Y]$	$[0, -x, 0]$	[7]
8	$[-Z, X, -Y]$	$[0, x, 0]$	[8]
9	$[Y, Z, X]$	$[0, 0, x]$	[9]
10	$[-Y, Z, -X]$	$[0, 0, -x]$	[10]
11	$[Y, -Z, -X]$	$[0, 0, -x]$	[11]
12	$[-Y, -Z, X]$	$[0, 0, x]$	[12]
13	$[Y, X, -Z]$	$[0, x, 0]$	[13]
14	$[-Y, -X, -Z]$	$[0, -x, 0]$	[14]
15	$[Y, -X, Z]$	$[0, -x, 0]$	[15]
16	$[-Y, X, Z]$	$[0, x, 0]$	[16]
17	$[X, Z, -Y]$	$[x, 0, 0]$	[17]
18	$[-X, Z, Y]$	$[-x, 0, 0]$	[18]
19	$[-X, -Z, -Y]$	$[-x, 0, 0]$	[19]
20	$[X, -Z, Y]$	$[x, 0, 0]$	[20]
21	$[Z, Y, -X]$	$[0, 0, -x]$	[21]
22	$[Z, -Y, X]$	$[0, 0, x]$	[22]

continued ...

Table 33

No.	vector	center	mapping
23	$[-Z, Y, X]$	$[0, 0, x]$	[23]
24	$[-Z, -Y, -X]$	$[0, 0, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, 0, 0]$	[25]
26	$[X, Y, -Z]$	$[x, 0, 0]$	[26]
27	$[X, -Y, Z]$	$[x, 0, 0]$	[27]
28	$[-X, Y, Z]$	$[-x, 0, 0]$	[28]
29	$[-Z, -X, -Y]$	$[0, -x, 0]$	[29]
30	$[-Z, X, Y]$	$[0, x, 0]$	[30]
31	$[Z, X, -Y]$	$[0, x, 0]$	[31]
32	$[Z, -X, Y]$	$[0, -x, 0]$	[32]
33	$[-Y, -Z, -X]$	$[0, 0, -x]$	[33]
34	$[Y, -Z, X]$	$[0, 0, x]$	[34]
35	$[-Y, Z, X]$	$[0, 0, x]$	[35]
36	$[Y, Z, -X]$	$[0, 0, -x]$	[36]
37	$[-Y, -X, Z]$	$[0, -x, 0]$	[37]
38	$[Y, X, Z]$	$[0, x, 0]$	[38]
39	$[-Y, X, -Z]$	$[0, x, 0]$	[39]
40	$[Y, -X, -Z]$	$[0, -x, 0]$	[40]
41	$[-X, -Z, Y]$	$[-x, 0, 0]$	[41]
42	$[X, -Z, -Y]$	$[x, 0, 0]$	[42]
43	$[X, Z, Y]$	$[x, 0, 0]$	[43]
44	$[-X, Z, -Y]$	$[-x, 0, 0]$	[44]
45	$[-Z, -Y, X]$	$[0, 0, x]$	[45]
46	$[-Z, Y, -X]$	$[0, 0, -x]$	[46]
47	$[Z, -Y, -X]$	$[0, 0, -x]$	[47]
48	$[Z, Y, X]$	$[0, 0, x]$	[48]

* Wyckoff site: 6f, site symmetry: 4m.m

Table 34: Wyckoff bond: 6a@6f, bond symmetry: 4m.m

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,4,17,20,26,27,42,43]
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,18,19,25,28,41,44]
3	$[0, X, 0]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,8,13,16,30,31,38,39]
4	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6,7,14,15,29,32,37,40]
5	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[9,12,22,23,34,35,45,48]
6	$[0, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10,11,21,24,33,36,46,47]

Table 35: Wyckoff bond: 12b@6f, bond symmetry: 2.mm

No.	vector	center	mapping
1	$[0, X, X]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,-4,-42,43]
2	$[0, -X, X]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[2,-3,41,-44]
3	$[X, 0, X]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,-8,38,-39]
4	$[X, 0, -X]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6,-7,-37,40]

continued ...

Table 35

No.	vector	center	mapping
5	$[X,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	`[9,-12,-45,48]`
6	$[-X,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	`[10,-11,46,-47]`
7	$[X,\ 0,\ -X]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[13,-16,-30,31]`
8	$[-X,\ 0,\ -X]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[14,-15,29,-32]`
9	$[0,\ X,\ -X]$	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,-20,26,-27]`
10	$[0,\ X,\ X]$	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,-19,-25,28]`
11	$[X,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	`[21,-24,-33,36]`
12	$[X,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	`[22,-23,34,-35]`

Table 36: Wyckoff bond: `12c@6f`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0,\ 0,\ X]$	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,-4,-26,27]`
2	$[0,\ 0,\ X]$	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,-3,-25,28]`
3	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[5,-8,-30,31]`
4	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[6,-7,-29,32]`
5	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	`[9,-12,-34,35]`
6	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	`[10,-11,-33,36]`
7	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[13,-16,-38,39]`
8	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[14,-15,-37,40]`
9	$[0,\ X,\ 0]$	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,-20,-42,43]`
10	$[0,\ X,\ 0]$	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,-19,-41,44]`
11	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	`[21,-24,-46,47]`
12	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	`[22,-23,-45,48]`

Table 37: Wyckoff bond: `24d@6f`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y,\ X,\ X]$	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,43]`
2	$[-Y,\ -X,\ X]$	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,41]`
3	$[-Y,\ X,\ -X]$	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,44]`
4	$[Y,\ -X,\ -X]$	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,42]`
5	$[X,\ Y,\ X]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[5,38]`
6	$[X,\ -Y,\ -X]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[6,40]`
7	$[-X,\ -Y,\ X]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[7,37]`
8	$[-X,\ Y,\ -X]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[8,39]`
9	$[X,\ X,\ Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	`[9,48]`
10	$[-X,\ X,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	`[10,46]`
11	$[X,\ -X,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ -x\right]$	`[11,47]`
12	$[-X,\ -X,\ Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x\right]$	`[12,45]`
13	$[X,\ Y,\ -X]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[13,31]`
14	$[-X,\ -Y,\ -X]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[14,29]`
15	$[X,\ -Y,\ X]$	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	`[15,32]`
16	$[-X,\ Y,\ X]$	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[16,30]`

continued ...

Table 37

No.	vector	center	mapping
17	$[Y, X, -X]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[17,26]
18	$[-Y, X, X]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[18,28]
19	$[-Y, -X, -X]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[19,25]
20	$[Y, -X, X]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[20,27]
21	$[X, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[21,36]
22	$[X, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[22,34]
23	$[-X, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[23,35]
24	$[-X, -X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[24,33]

Table 38: Wyckoff bond: 24e@6f, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,26]
2	$[-X, -Y, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[2,25]
3	$[-X, Y, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[3,28]
4	$[X, -Y, 0]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[4,27]
5	$[0, X, Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,30]
6	$[0, -X, -Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6,29]
7	$[0, -X, Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[7,32]
8	$[0, X, -Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[8,31]
9	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[9,34]
10	$[-Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10,33]
11	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[11,36]
12	$[-Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[12,35]
13	$[Y, X, 0]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[13,38]
14	$[-Y, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[14,37]
15	$[Y, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[15,40]
16	$[-Y, X, 0]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[16,39]
17	$[X, 0, -Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[17,42]
18	$[-X, 0, Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[18,41]
19	$[-X, 0, -Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[19,44]
20	$[X, 0, Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[20,43]
21	$[0, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[21,46]
22	$[0, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[22,45]
23	$[0, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[23,48]
24	$[0, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[24,47]

Table 39: Wyckoff bond: 24f@6f, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6,-7]

continued ...

Table 39

No.	vector	center	mapping
5	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10,-11]
7	$[X, 0, -Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[13,-16]
8	$[-X, 0, -Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[17,-20]
10	$[0, Y, X]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[46,-47]

Table 40: Wyckoff bond: 48g@6f, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[17]
18	$[-X, Z, Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[21]

continued ...

Table 40

No.	vector	center	mapping
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[42]
43	$[X, Z, Y]$	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[48]

* Wyckoff site: 8g, site symmetry: .3m

Table 41: Wyckoff bond: 8a@8g, bond symmetry: .3m

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9,38,43,48]
2	$[-X, -X, X]$	$[-x, -x, x]$	[2,7,12,37,41,45]
3	$[-X, X, -X]$	$[-x, x, -x]$	[3,8,10,39,44,46]
4	$[X, -X, -X]$	$[x, -x, -x]$	[4,6,11,40,42,47]
5	$[X, X, -X]$	$[x, x, -x]$	[13,17,21,26,31,36]
6	$[-X, -X, -X]$	$[-x, -x, -x]$	[14,19,24,25,29,33]
7	$[X, -X, X]$	$[x, -x, x]$	[15,20,22,27,32,34]
8	$[-X, X, X]$	$[-x, x, x]$	[16,18,23,28,30,35]

Table 42: Wyckoff bond: `24b@8g`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, x]$	`[1,38]`
2	$[-X, -X, Y]$	$[-x, -x, x]$	`[2,37]`
3	$[-X, X, -Y]$	$[-x, x, -x]$	`[3,39]`
4	$[X, -X, -Y]$	$[x, -x, -x]$	`[4,40]`
5	$[Y, X, X]$	$[x, x, x]$	`[5,48]`
6	$[Y, -X, -X]$	$[x, -x, -x]$	`[6,47]`
7	$[-Y, -X, X]$	$[-x, -x, x]$	`[7,45]`
8	$[-Y, X, -X]$	$[-x, x, -x]$	`[8,46]`
9	$[X, Y, X]$	$[x, x, x]$	`[9,43]`
10	$[-X, Y, -X]$	$[-x, x, -x]$	`[10,44]`
11	$[X, -Y, -X]$	$[x, -x, -x]$	`[11,42]`
12	$[-X, -Y, X]$	$[-x, -x, x]$	`[12,41]`
13	$[X, X, -Y]$	$[x, x, -x]$	`[13,26]`
14	$[-X, -X, -Y]$	$[-x, -x, -x]$	`[14,25]`
15	$[X, -X, Y]$	$[x, -x, x]$	`[15,27]`
16	$[-X, X, Y]$	$[-x, x, x]$	`[16,28]`
17	$[X, Y, -X]$	$[x, x, -x]$	`[17,36]`
18	$[-X, Y, X]$	$[-x, x, x]$	`[18,35]`
19	$[-X, -Y, -X]$	$[-x, -x, -x]$	`[19,33]`
20	$[X, -Y, X]$	$[x, -x, x]$	`[20,34]`
21	$[Y, X, -X]$	$[x, x, -x]$	`[21,31]`
22	$[Y, -X, X]$	$[x, -x, x]$	`[22,32]`
23	$[-Y, X, X]$	$[-x, x, x]$	`[23,30]`
24	$[-Y, -X, -X]$	$[-x, -x, -x]$	`[24,29]`

Table 43: Wyckoff bond: `24c@8g`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, x]$	`[1,-38]`
2	$[-X, X, 0]$	$[-x, -x, x]$	`[2,-37]`
3	$[-X, -X, 0]$	$[-x, x, -x]$	`[3,-39]`
4	$[X, X, 0]$	$[x, -x, -x]$	`[4,-40]`
5	$[0, X, -X]$	$[x, x, x]$	`[5,-48]`
6	$[0, -X, X]$	$[x, -x, -x]$	`[6,-47]`
7	$[0, -X, -X]$	$[-x, -x, x]$	`[7,-45]`
8	$[0, X, X]$	$[-x, x, -x]$	`[8,-46]`
9	$[-X, 0, X]$	$[x, x, x]$	`[9,-43]`
10	$[X, 0, -X]$	$[-x, x, -x]$	`[10,-44]`
11	$[-X, 0, -X]$	$[x, -x, -x]$	`[11,-42]`
12	$[X, 0, X]$	$[-x, -x, x]$	`[12,-41]`
13	$[-X, X, 0]$	$[x, x, -x]$	`[13,-26]`
14	$[X, -X, 0]$	$[-x, -x, -x]$	`[14,-25]`
15	$[-X, -X, 0]$	$[x, -x, x]$	`[15,-27]`
16	$[X, X, 0]$	$[-x, x, x]$	`[16,-28]`
17	$[X, 0, X]$	$[x, x, -x]$	`[17,-36]`

continued ...

Table 43

No.	vector	center	mapping
18	$[-X, 0, -X]$	$[-x, x, x]$	[18,-35]
19	$[-X, 0, X]$	$[-x, -x, -x]$	[19,-33]
20	$[X, 0, -X]$	$[x, -x, x]$	[20,-34]
21	$[0, -X, -X]$	$[x, x, -x]$	[21,-31]
22	$[0, X, X]$	$[x, -x, x]$	[22,-32]
23	$[0, -X, X]$	$[-x, x, x]$	[23,-30]
24	$[0, X, -X]$	$[-x, -x, -x]$	[24,-29]

Table 44: Wyckoff bond: 48d@8g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$[-x, -x, x]$	[2]
3	$[-X, Y, -Z]$	$[-x, x, -x]$	[3]
4	$[X, -Y, -Z]$	$[x, -x, -x]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$[x, -x, -x]$	[6]
7	$[-Z, -X, Y]$	$[-x, -x, x]$	[7]
8	$[-Z, X, -Y]$	$[-x, x, -x]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$[-x, x, -x]$	[10]
11	$[Y, -Z, -X]$	$[x, -x, -x]$	[11]
12	$[-Y, -Z, X]$	$[-x, -x, x]$	[12]
13	$[Y, X, -Z]$	$[x, x, -x]$	[13]
14	$[-Y, -X, -Z]$	$[-x, -x, -x]$	[14]
15	$[Y, -X, Z]$	$[x, -x, x]$	[15]
16	$[-Y, X, Z]$	$[-x, x, x]$	[16]
17	$[X, Z, -Y]$	$[x, x, -x]$	[17]
18	$[-X, Z, Y]$	$[-x, x, x]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -x, -x]$	[19]
20	$[X, -Z, Y]$	$[x, -x, x]$	[20]
21	$[Z, Y, -X]$	$[x, x, -x]$	[21]
22	$[Z, -Y, X]$	$[x, -x, x]$	[22]
23	$[-Z, Y, X]$	$[-x, x, x]$	[23]
24	$[-Z, -Y, -X]$	$[-x, -x, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -x, -x]$	[25]
26	$[X, Y, -Z]$	$[x, x, -x]$	[26]
27	$[X, -Y, Z]$	$[x, -x, x]$	[27]
28	$[-X, Y, Z]$	$[-x, x, x]$	[28]
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[29]
30	$[-Z, X, Y]$	$[-x, x, x]$	[30]
31	$[Z, X, -Y]$	$[x, x, -x]$	[31]
32	$[Z, -X, Y]$	$[x, -x, x]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[33]
34	$[Y, -Z, X]$	$[x, -x, x]$	[34]

continued ...

Table 44

No.	vector	center	mapping
35	$[-Y, Z, X]$	$[-x, x, x]$	[35]
36	$[Y, Z, -X]$	$[x, x, -x]$	[36]
37	$[-Y, -X, Z]$	$[-x, -x, x]$	[37]
38	$[Y, X, Z]$	$[x, x, x]$	[38]
39	$[-Y, X, -Z]$	$[-x, x, -x]$	[39]
40	$[Y, -X, -Z]$	$[x, -x, -x]$	[40]
41	$[-X, -Z, Y]$	$[-x, -x, x]$	[41]
42	$[X, -Z, -Y]$	$[x, -x, -x]$	[42]
43	$[X, Z, Y]$	$[x, x, x]$	[43]
44	$[-X, Z, -Y]$	$[-x, x, -x]$	[44]
45	$[-Z, -Y, X]$	$[-x, -x, x]$	[45]
46	$[-Z, Y, -X]$	$[-x, x, -x]$	[46]
47	$[Z, -Y, -X]$	$[x, -x, -x]$	[47]
48	$[Z, Y, X]$	$[x, x, x]$	[48]

* Wyckoff site: 12h, site symmetry: mm2..

Table 45: Wyckoff bond: 12a@12h, bond symmetry: mm2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, 0\right]$	[1,4,26,27]
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	[2,3,25,28]
3	$[0, X, 0]$	$\left[0, x, \frac{1}{2}\right]$	[5,8,30,31]
4	$[0, -X, 0]$	$\left[0, -x, \frac{1}{2}\right]$	[6,7,29,32]
5	$[0, 0, X]$	$\left[\frac{1}{2}, 0, x\right]$	[9,12,34,35]
6	$[0, 0, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[10,11,33,36]
7	$[0, X, 0]$	$\left[\frac{1}{2}, x, 0\right]$	[13,16,38,39]
8	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, 0\right]$	[14,15,37,40]
9	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	[17,20,42,43]
10	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	[18,19,41,44]
11	$[0, 0, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[21,24,46,47]
12	$[0, 0, X]$	$\left[0, \frac{1}{2}, x\right]$	[22,23,45,48]

Table 46: Wyckoff bond: 12b@12h, bond symmetry: mm2..

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[x, \frac{1}{2}, 0\right]$	[1,-4,-26,27]
2	$[0, 0, X]$	$\left[-x, \frac{1}{2}, 0\right]$	[2,-3,-25,28]
3	$[X, 0, 0]$	$\left[0, x, \frac{1}{2}\right]$	[5,-8,-30,31]
4	$[X, 0, 0]$	$\left[0, -x, \frac{1}{2}\right]$	[6,-7,-29,32]
5	$[0, X, 0]$	$\left[\frac{1}{2}, 0, x\right]$	[9,-12,-34,35]
6	$[0, X, 0]$	$\left[\frac{1}{2}, 0, -x\right]$	[10,-11,-33,36]
7	$[0, 0, -X]$	$\left[\frac{1}{2}, x, 0\right]$	[13,-16,-38,39]
8	$[0, 0, -X]$	$\left[\frac{1}{2}, -x, 0\right]$	[14,-15,-37,40]
9	$[0, X, 0]$	$\left[x, 0, \frac{1}{2}\right]$	[17,-20,-42,43]
10	$[0, X, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	[18,-19,-41,44]

continued ...

Table 46

No.	vector	center	mapping
11	$[X, 0, 0]$	$\left[0, \frac{1}{2}, -x\right]$	`[21,-24,-46,47]`
12	$[X, 0, 0]$	$\left[0, \frac{1}{2}, x\right]$	`[22,-23,-45,48]`

Table 47: Wyckoff bond: `12c@12h`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,-4,26,-27]`
2	$[0, -X, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,-3,25,-28]`
3	$[0, 0, X]$	$\left[0, x, \frac{1}{2}\right]$	`[5,-8,30,-31]`
4	$[0, 0, -X]$	$\left[0, -x, \frac{1}{2}\right]$	`[6,-7,29,-32]`
5	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, x\right]$	`[9,-12,34,-35]`
6	$[-X, 0, 0]$	$\left[\frac{1}{2}, 0, -x\right]$	`[10,-11,33,-36]`
7	$[X, 0, 0]$	$\left[\frac{1}{2}, x, 0\right]$	`[13,-16,38,-39]`
8	$[-X, 0, 0]$	$\left[\frac{1}{2}, -x, 0\right]$	`[14,-15,37,-40]`
9	$[0, 0, -X]$	$\left[x, 0, \frac{1}{2}\right]$	`[17,-20,42,-43]`
10	$[0, 0, X]$	$\left[-x, 0, \frac{1}{2}\right]$	`[18,-19,41,-44]`
11	$[0, X, 0]$	$\left[0, \frac{1}{2}, -x\right]$	`[21,-24,46,-47]`
12	$[0, -X, 0]$	$\left[0, \frac{1}{2}, x\right]$	`[22,-23,45,-48]`

Table 48: Wyckoff bond: `24d@12h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,26]`
2	$[-X, -Y, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,25]`
3	$[-X, Y, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	`[3,28]`
4	$[X, -Y, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[4,27]`
5	$[0, X, Y]$	$\left[0, x, \frac{1}{2}\right]$	`[5,30]`
6	$[0, -X, -Y]$	$\left[0, -x, \frac{1}{2}\right]$	`[6,29]`
7	$[0, -X, Y]$	$\left[0, -x, \frac{1}{2}\right]$	`[7,32]`
8	$[0, X, -Y]$	$\left[0, x, \frac{1}{2}\right]$	`[8,31]`
9	$[Y, 0, X]$	$\left[\frac{1}{2}, 0, x\right]$	`[9,34]`
10	$[-Y, 0, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	`[10,33]`
11	$[Y, 0, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	`[11,36]`
12	$[-Y, 0, X]$	$\left[\frac{1}{2}, 0, x\right]$	`[12,35]`
13	$[Y, X, 0]$	$\left[\frac{1}{2}, x, 0\right]$	`[13,38]`
14	$[-Y, -X, 0]$	$\left[\frac{1}{2}, -x, 0\right]$	`[14,37]`
15	$[Y, -X, 0]$	$\left[\frac{1}{2}, -x, 0\right]$	`[15,40]`
16	$[-Y, X, 0]$	$\left[\frac{1}{2}, x, 0\right]$	`[16,39]`
17	$[X, 0, -Y]$	$\left[x, 0, \frac{1}{2}\right]$	`[17,42]`
18	$[-X, 0, Y]$	$\left[-x, 0, \frac{1}{2}\right]$	`[18,41]`
19	$[-X, 0, -Y]$	$\left[-x, 0, \frac{1}{2}\right]$	`[19,44]`
20	$[X, 0, Y]$	$\left[x, 0, \frac{1}{2}\right]$	`[20,43]`
21	$[0, Y, -X]$	$\left[0, \frac{1}{2}, -x\right]$	`[21,46]`
22	$[0, -Y, X]$	$\left[0, \frac{1}{2}, x\right]$	`[22,45]`

continued ...

Table 48

No.	vector	center	mapping
23	$[0,\ Y,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[23,48]`
24	$[0,\ -Y,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[24,47]`

Table 49: Wyckoff bond: `24e@12h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[Y,\ 0,\ X]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,27]`
2	$[-Y,\ 0,\ X]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,28]`
3	$[-Y,\ 0,\ -X]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,25]`
4	$[Y,\ 0,\ -X]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4,26]`
5	$[X,\ Y,\ 0]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,31]`
6	$[X,\ -Y,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,32]`
7	$[-X,\ -Y,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[7,29]`
8	$[-X,\ Y,\ 0]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[8,30]`
9	$[0,\ X,\ Y]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,35]`
10	$[0,\ X,\ -Y]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,36]`
11	$[0,\ -X,\ -Y]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[11,33]`
12	$[0,\ -X,\ Y]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[12,34]`
13	$[0,\ Y,\ -X]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[13,39]`
14	$[0,\ -Y,\ -X]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[14,40]`
15	$[0,\ -Y,\ X]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[15,37]`
16	$[0,\ Y,\ X]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[16,38]`
17	$[Y,\ X,\ 0]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[17,43]`
18	$[-Y,\ X,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[18,44]`
19	$[-Y,\ -X,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[19,41]`
20	$[Y,\ -X,\ 0]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[20,42]`
21	$[X,\ 0,\ -Y]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[21,47]`
22	$[X,\ 0,\ Y]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[22,48]`
23	$[-X,\ 0,\ Y]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[23,45]`
24	$[-X,\ 0,\ -Y]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[24,46]`

Table 50: Wyckoff bond: `24f@12h`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,-8]`
4	$[Y,\ 0,\ -X]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,-7]`
5	$[X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,-12]`
6	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,-11]`
7	$[X,\ 0,\ -Y]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[13,-16]`
8	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[14,-15]`
9	$[0,\ Y,\ -X]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[17,-20]`
10	$[0,\ Y,\ X]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[18,-19]`

continued ...

Table 50

No.	vector	center	mapping
11	$[Y, X, 0]$	$\left[0, \frac{1}{2}, -x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, x\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[-x, \frac{1}{2}, 0\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[x, \frac{1}{2}, 0\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[0, -x, \frac{1}{2}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[0, x, \frac{1}{2}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{1}{2}, 0, -x\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, x\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{1}{2}, -x, 0\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{2}, x, 0\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[-x, 0, \frac{1}{2}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x, 0, \frac{1}{2}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, x\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[0, \frac{1}{2}, -x\right]$	[46,-47]

Table 51: Wyckoff bond: 48g@12h, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	[4]
5	$[Z, X, Y]$	$\left[0, x, \frac{1}{2}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[0, -x, \frac{1}{2}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[0, -x, \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[0, x, \frac{1}{2}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, x, 0\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, -x, 0\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, -x, 0\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, x, 0\right]$	[16]
17	$[X, Z, -Y]$	$\left[x, 0, \frac{1}{2}\right]$	[17]
18	$[-X, Z, Y]$	$\left[-x, 0, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[-x, 0, \frac{1}{2}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x, 0, \frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[21]
22	$[Z, -Y, X]$	$\left[0, \frac{1}{2}, x\right]$	[22]
23	$[-Z, Y, X]$	$\left[0, \frac{1}{2}, x\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[25]
26	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	[26]
27	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	[27]

continued ...

Table 51

No.	vector	center	mapping
28	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[0, -x, \frac{1}{2}\right]$	[29]
30	$[-Z, X, Y]$	$\left[0, x, \frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[0, x, \frac{1}{2}\right]$	[31]
32	$[Z, -X, Y]$	$\left[0, -x, \frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}, -x, 0\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{2}, x, 0\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}, x, 0\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{2}, -x, 0\right]$	[40]
41	$[-X, -Z, Y]$	$\left[-x, 0, \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x, 0, \frac{1}{2}\right]$	[42]
43	$[X, Z, Y]$	$\left[x, 0, \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[-x, 0, \frac{1}{2}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[0, \frac{1}{2}, x\right]$	[45]
46	$[-Z, Y, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[47]
48	$[Z, Y, X]$	$\left[0, \frac{1}{2}, x\right]$	[48]

* Wyckoff site: 12i, site symmetry: m.m2

Table 52: Wyckoff bond: 12a@12i, bond symmetry: m.m2

No.	vector	center	mapping
1	$[0, X, X]$	$[0, y, y]$	[1,18,28,43]
2	$[0, -X, X]$	$[0, -y, y]$	[2,20,27,41]
3	$[0, X, -X]$	$[0, y, -y]$	[3,17,26,44]
4	$[0, -X, -X]$	$[0, -y, -y]$	[4,19,25,42]
5	$[X, 0, X]$	$[y, 0, y]$	[5,15,32,38]
6	$[X, 0, -X]$	$[y, 0, -y]$	[6,13,31,40]
7	$[-X, 0, X]$	$[-y, 0, y]$	[7,16,30,37]
8	$[-X, 0, -X]$	$[-y, 0, -y]$	[8,14,29,39]
9	$[X, X, 0]$	$[y, y, 0]$	[9,21,36,48]
10	$[-X, X, 0]$	$[-y, y, 0]$	[10,23,35,46]
11	$[X, -X, 0]$	$[y, -y, 0]$	[11,22,34,47]
12	$[-X, -X, 0]$	$[-y, -y, 0]$	[12,24,33,45]

Table 53: Wyckoff bond: 12b@12i, bond symmetry: m.m2

No.	vector	center	mapping
1	$[0, X, -X]$	$[0, y, y]$	[1,-18,28,-43]
2	$[0, -X, -X]$	$[0, -y, y]$	[2,-20,27,-41]
3	$[0, X, X]$	$[0, y, -y]$	[3,-17,26,-44]

continued ...

Table 53

No.	vector	center	mapping
4	$[0, -X, X]$	$[0, -y, -y]$	`[4,-19,25,-42]`
5	$[-X, 0, X]$	$[y, 0, y]$	`[5,-15,32,-38]`
6	$[-X, 0, -X]$	$[y, 0, -y]$	`[6,-13,31,-40]`
7	$[X, 0, X]$	$[-y, 0, y]$	`[7,-16,30,-37]`
8	$[X, 0, -X]$	$[-y, 0, -y]$	`[8,-14,29,-39]`
9	$[X, -X, 0]$	$[y, y, 0]$	`[9,-21,36,-48]`
10	$[-X, -X, 0]$	$[-y, y, 0]$	`[10,-23,35,-46]`
11	$[X, X, 0]$	$[y, -y, 0]$	`[11,-22,34,-47]`
12	$[-X, X, 0]$	$[-y, -y, 0]$	`[12,-24,33,-45]`

Table 54: Wyckoff bond: `12c@12i`, bond symmetry: `m.m2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, y]$	`[1,-18,-28,43]`
2	$[-X, 0, 0]$	$[0, -y, y]$	`[2,-20,-27,41]`
3	$[-X, 0, 0]$	$[0, y, -y]$	`[3,-17,-26,44]`
4	$[X, 0, 0]$	$[0, -y, -y]$	`[4,-19,-25,42]`
5	$[0, X, 0]$	$[y, 0, y]$	`[5,-15,-32,38]`
6	$[0, -X, 0]$	$[y, 0, -y]$	`[6,-13,-31,40]`
7	$[0, -X, 0]$	$[-y, 0, y]$	`[7,-16,-30,37]`
8	$[0, X, 0]$	$[-y, 0, -y]$	`[8,-14,-29,39]`
9	$[0, 0, X]$	$[y, y, 0]$	`[9,-21,-36,48]`
10	$[0, 0, -X]$	$[-y, y, 0]$	`[10,-23,-35,46]`
11	$[0, 0, -X]$	$[y, -y, 0]$	`[11,-22,-34,47]`
12	$[0, 0, X]$	$[-y, -y, 0]$	`[12,-24,-33,45]`

Table 55: Wyckoff bond: `24d@12i`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y, X, X]$	$[0, y, y]$	`[1,43]`
2	$[-Y, -X, X]$	$[0, -y, y]$	`[2,41]`
3	$[-Y, X, -X]$	$[0, y, -y]$	`[3,44]`
4	$[Y, -X, -X]$	$[0, -y, -y]$	`[4,42]`
5	$[X, Y, X]$	$[y, 0, y]$	`[5,38]`
6	$[X, -Y, -X]$	$[y, 0, -y]$	`[6,40]`
7	$[-X, -Y, X]$	$[-y, 0, y]$	`[7,37]`
8	$[-X, Y, -X]$	$[-y, 0, -y]$	`[8,39]`
9	$[X, X, Y]$	$[y, y, 0]$	`[9,48]`
10	$[-X, X, -Y]$	$[-y, y, 0]$	`[10,46]`
11	$[X, -X, -Y]$	$[y, -y, 0]$	`[11,47]`
12	$[-X, -X, Y]$	$[-y, -y, 0]$	`[12,45]`
13	$[X, Y, -X]$	$[y, 0, -y]$	`[13,31]`
14	$[-X, -Y, -X]$	$[-y, 0, -y]$	`[14,29]`
15	$[X, -Y, X]$	$[y, 0, y]$	`[15,32]`

continued ...

Table 55

No.	vector	center	mapping
16	$[-X, Y, X]$	$[-y, 0, y]$	`[16,30]`
17	$[Y, X, -X]$	$[0, y, -y]$	`[17,26]`
18	$[-Y, X, X]$	$[0, y, y]$	`[18,28]`
19	$[-Y, -X, -X]$	$[0, -y, -y]$	`[19,25]`
20	$[Y, -X, X]$	$[0, -y, y]$	`[20,27]`
21	$[X, X, -Y]$	$[y, y, 0]$	`[21,36]`
22	$[X, -X, Y]$	$[y, -y, 0]$	`[22,34]`
23	$[-X, X, Y]$	$[-y, y, 0]$	`[23,35]`
24	$[-X, -X, -Y]$	$[-y, -y, 0]$	`[24,33]`

Table 56: Wyckoff bond: `24e@12i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$[0, y, y]$	`[1,-18]`
2	$[-Y, -X, -X]$	$[0, -y, y]$	`[2,-20]`
3	$[-Y, X, X]$	$[0, y, -y]$	`[3,-17]`
4	$[Y, -X, X]$	$[0, -y, -y]$	`[4,-19]`
5	$[-X, Y, X]$	$[y, 0, y]$	`[5,-15]`
6	$[-X, -Y, -X]$	$[y, 0, -y]$	`[6,-13]`
7	$[X, -Y, X]$	$[-y, 0, y]$	`[7,-16]`
8	$[X, Y, -X]$	$[-y, 0, -y]$	`[8,-14]`
9	$[X, -X, Y]$	$[y, y, 0]$	`[9,-21]`
10	$[-X, -X, -Y]$	$[-y, y, 0]$	`[10,-23]`
11	$[X, X, -Y]$	$[y, -y, 0]$	`[11,-22]`
12	$[-X, X, Y]$	$[-y, -y, 0]$	`[12,-24]`
13	$[-Y, -X, X]$	$[0, -y, -y]$	`[25,-42]`
14	$[Y, X, X]$	$[0, y, -y]$	`[26,-44]`
15	$[Y, -X, -X]$	$[0, -y, y]$	`[27,-41]`
16	$[-Y, X, -X]$	$[0, y, y]$	`[28,-43]`
17	$[X, -Y, -X]$	$[-y, 0, -y]$	`[29,-39]`
18	$[X, Y, X]$	$[-y, 0, y]$	`[30,-37]`
19	$[-X, Y, -X]$	$[y, 0, -y]$	`[31,-40]`
20	$[-X, -Y, X]$	$[y, 0, y]$	`[32,-38]`
21	$[-X, X, -Y]$	$[-y, -y, 0]$	`[33,-45]`
22	$[X, X, Y]$	$[y, -y, 0]$	`[34,-47]`
23	$[-X, -X, Y]$	$[-y, y, 0]$	`[35,-46]`
24	$[X, -X, -Y]$	$[y, y, 0]$	`[36,-48]`

Table 57: Wyckoff bond: `24f@12i`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, y, y]$	`[1,28]`
2	$[0, -X, Y]$	$[0, -y, y]$	`[2,27]`
3	$[0, X, -Y]$	$[0, y, -y]$	`[3,26]`

continued ...

Table 57

No.	vector	center	mapping
4	$[0, -X, -Y]$	$[0, -y, -y]$	[4,25]
5	$[Y, 0, X]$	$[y, 0, y]$	[5,32]
6	$[Y, 0, -X]$	$[y, 0, -y]$	[6,31]
7	$[-Y, 0, X]$	$[-y, 0, y]$	[7,30]
8	$[-Y, 0, -X]$	$[-y, 0, -y]$	[8,29]
9	$[X, Y, 0]$	$[y, y, 0]$	[9,36]
10	$[-X, Y, 0]$	$[-y, y, 0]$	[10,35]
11	$[X, -Y, 0]$	$[y, -y, 0]$	[11,34]
12	$[-X, -Y, 0]$	$[-y, -y, 0]$	[12,33]
13	$[X, 0, -Y]$	$[y, 0, -y]$	[13,40]
14	$[-X, 0, -Y]$	$[-y, 0, -y]$	[14,39]
15	$[X, 0, Y]$	$[y, 0, y]$	[15,38]
16	$[-X, 0, Y]$	$[-y, 0, y]$	[16,37]
17	$[0, Y, -X]$	$[0, y, -y]$	[17,44]
18	$[0, Y, X]$	$[0, y, y]$	[18,43]
19	$[0, -Y, -X]$	$[0, -y, -y]$	[19,42]
20	$[0, -Y, X]$	$[0, -y, y]$	[20,41]
21	$[Y, X, 0]$	$[y, y, 0]$	[21,48]
22	$[Y, -X, 0]$	$[y, -y, 0]$	[22,47]
23	$[-Y, X, 0]$	$[-y, y, 0]$	[23,46]
24	$[-Y, -X, 0]$	$[-y, -y, 0]$	[24,45]

Table 58: Wyckoff bond: 48g@12i, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, y]$	[1]
2	$[-X, -Y, Z]$	$[0, -y, y]$	[2]
3	$[-X, Y, -Z]$	$[0, y, -y]$	[3]
4	$[X, -Y, -Z]$	$[0, -y, -y]$	[4]
5	$[Z, X, Y]$	$[y, 0, y]$	[5]
6	$[Z, -X, -Y]$	$[y, 0, -y]$	[6]
7	$[-Z, -X, Y]$	$[-y, 0, y]$	[7]
8	$[-Z, X, -Y]$	$[-y, 0, -y]$	[8]
9	$[Y, Z, X]$	$[y, y, 0]$	[9]
10	$[-Y, Z, -X]$	$[-y, y, 0]$	[10]
11	$[Y, -Z, -X]$	$[y, -y, 0]$	[11]
12	$[-Y, -Z, X]$	$[-y, -y, 0]$	[12]
13	$[Y, X, -Z]$	$[y, 0, -y]$	[13]
14	$[-Y, -X, -Z]$	$[-y, 0, -y]$	[14]
15	$[Y, -X, Z]$	$[y, 0, y]$	[15]
16	$[-Y, X, Z]$	$[-y, 0, y]$	[16]
17	$[X, Z, -Y]$	$[0, y, -y]$	[17]
18	$[-X, Z, Y]$	$[0, y, y]$	[18]
19	$[-X, -Z, -Y]$	$[0, -y, -y]$	[19]
20	$[X, -Z, Y]$	$[0, -y, y]$	[20]

continued ...

Table 58

No.	vector	center	mapping
21	$[Z, Y, -X]$	$[y, y, 0]$	[21]
22	$[Z, -Y, X]$	$[y, -y, 0]$	[22]
23	$[-Z, Y, X]$	$[-y, y, 0]$	[23]
24	$[-Z, -Y, -X]$	$[-y, -y, 0]$	[24]
25	$[-X, -Y, -Z]$	$[0, -y, -y]$	[25]
26	$[X, Y, -Z]$	$[0, y, -y]$	[26]
27	$[X, -Y, Z]$	$[0, -y, y]$	[27]
28	$[-X, Y, Z]$	$[0, y, y]$	[28]
29	$[-Z, -X, -Y]$	$[-y, 0, -y]$	[29]
30	$[-Z, X, Y]$	$[-y, 0, y]$	[30]
31	$[Z, X, -Y]$	$[y, 0, -y]$	[31]
32	$[Z, -X, Y]$	$[y, 0, y]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -y, 0]$	[33]
34	$[Y, -Z, X]$	$[y, -y, 0]$	[34]
35	$[-Y, Z, X]$	$[-y, y, 0]$	[35]
36	$[Y, Z, -X]$	$[y, y, 0]$	[36]
37	$[-Y, -X, Z]$	$[-y, 0, y]$	[37]
38	$[Y, X, Z]$	$[y, 0, y]$	[38]
39	$[-Y, X, -Z]$	$[-y, 0, -y]$	[39]
40	$[Y, -X, -Z]$	$[y, 0, -y]$	[40]
41	$[-X, -Z, Y]$	$[0, -y, y]$	[41]
42	$[X, -Z, -Y]$	$[0, -y, -y]$	[42]
43	$[X, Z, Y]$	$[0, y, y]$	[43]
44	$[-X, Z, -Y]$	$[0, y, -y]$	[44]
45	$[-Z, -Y, X]$	$[-y, -y, 0]$	[45]
46	$[-Z, Y, -X]$	$[-y, y, 0]$	[46]
47	$[Z, -Y, -X]$	$[y, -y, 0]$	[47]
48	$[Z, Y, X]$	$[y, y, 0]$	[48]

* Wyckoff site: 12j, site symmetry: m.m2

Table 59: Wyckoff bond: 12a@12j, bond symmetry: m.m2

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{2}, y, y\right]$	[1,18,28,43]
2	$[0, -X, X]$	$\left[\frac{1}{2}, -y, y\right]$	[2,20,27,41]
3	$[0, X, -X]$	$\left[\frac{1}{2}, y, -y\right]$	[3,17,26,44]
4	$[0, -X, -X]$	$\left[\frac{1}{2}, -y, -y\right]$	[4,19,25,42]
5	$[X, 0, X]$	$\left[y, \frac{1}{2}, y\right]$	[5,15,32,38]
6	$[X, 0, -X]$	$\left[y, \frac{1}{2}, -y\right]$	[6,13,31,40]
7	$[-X, 0, X]$	$\left[-y, \frac{1}{2}, y\right]$	[7,16,30,37]
8	$[-X, 0, -X]$	$\left[-y, \frac{1}{2}, -y\right]$	[8,14,29,39]
9	$[X, X, 0]$	$\left[y, y, \frac{1}{2}\right]$	[9,21,36,48]
10	$[-X, X, 0]$	$\left[-y, y, \frac{1}{2}\right]$	[10,23,35,46]
11	$[X, -X, 0]$	$\left[y, -y, \frac{1}{2}\right]$	[11,22,34,47]
12	$[-X, -X, 0]$	$\left[-y, -y, \frac{1}{2}\right]$	[12,24,33,45]

Table 60: Wyckoff bond: `12b@12j`, bond symmetry: `m.m2`

No.	vector	center	mapping
1	$[0,\ X,\ -X]$	$\left[\frac{1}{2},\ y,\ y\right]$	`[1,-18,28,-43]`
2	$[0,\ -X,\ -X]$	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2,-20,27,-41]`
3	$[0,\ X,\ X]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3,-17,26,-44]`
4	$[0,\ -X,\ X]$	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4,-19,25,-42]`
5	$[-X,\ 0,\ X]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[5,-15,32,-38]`
6	$[-X,\ 0,\ -X]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6,-13,31,-40]`
7	$[X,\ 0,\ X]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7,-16,30,-37]`
8	$[X,\ 0,\ -X]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8,-14,29,-39]`
9	$[X,\ -X,\ 0]$	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9,-21,36,-48]`
10	$[-X,\ -X,\ 0]$	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10,-23,35,-46]`
11	$[X,\ X,\ 0]$	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11,-22,34,-47]`
12	$[-X,\ X,\ 0]$	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12,-24,33,-45]`

Table 61: Wyckoff bond: `12c@12j`, bond symmetry: `m.m2`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ y,\ y\right]$	`[1,-18,-28,43]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2,-20,-27,41]`
3	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3,-17,-26,44]`
4	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4,-19,-25,42]`
5	$[0,\ X,\ 0]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[5,-15,-32,38]`
6	$[0,\ -X,\ 0]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6,-13,-31,40]`
7	$[0,\ -X,\ 0]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7,-16,-30,37]`
8	$[0,\ X,\ 0]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8,-14,-29,39]`
9	$[0,\ 0,\ X]$	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9,-21,-36,48]`
10	$[0,\ 0,\ -X]$	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10,-23,-35,46]`
11	$[0,\ 0,\ -X]$	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11,-22,-34,47]`
12	$[0,\ 0,\ X]$	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12,-24,-33,45]`

Table 62: Wyckoff bond: `24d@12j`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[Y,\ X,\ X]$	$\left[\frac{1}{2},\ y,\ y\right]$	`[1,43]`
2	$[-Y,\ -X,\ X]$	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2,41]`
3	$[-Y,\ X,\ -X]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3,44]`
4	$[Y,\ -X,\ -X]$	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4,42]`
5	$[X,\ Y,\ X]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[5,38]`
6	$[X,\ -Y,\ -X]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6,40]`
7	$[-X,\ -Y,\ X]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7,37]`
8	$[-X,\ Y,\ -X]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8,39]`
9	$[X,\ X,\ Y]$	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9,48]`
10	$[-X,\ X,\ -Y]$	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10,46]`
11	$[X,\ -X,\ -Y]$	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11,47]`
12	$[-X,\ -X,\ Y]$	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12,45]`

continued ...

Table 62

No.	vector	center	mapping
13	$[X, Y, -X]$	$\left[y, \frac{1}{2}, -y\right]$	[13,31]
14	$[-X, -Y, -X]$	$\left[-y, \frac{1}{2}, -y\right]$	[14,29]
15	$[X, -Y, X]$	$\left[y, \frac{1}{2}, y\right]$	[15,32]
16	$[-X, Y, X]$	$\left[-y, \frac{1}{2}, y\right]$	[16,30]
17	$[Y, X, -X]$	$\left[\frac{1}{2}, y, -y\right]$	[17,26]
18	$[-Y, X, X]$	$\left[\frac{1}{2}, y, y\right]$	[18,28]
19	$[-Y, -X, -X]$	$\left[\frac{1}{2}, -y, -y\right]$	[19,25]
20	$[Y, -X, X]$	$\left[\frac{1}{2}, -y, y\right]$	[20,27]
21	$[X, X, -Y]$	$\left[y, y, \frac{1}{2}\right]$	[21,36]
22	$[X, -X, Y]$	$\left[y, -y, \frac{1}{2}\right]$	[22,34]
23	$[-X, X, Y]$	$\left[-y, y, \frac{1}{2}\right]$	[23,35]
24	$[-X, -X, -Y]$	$\left[-y, -y, \frac{1}{2}\right]$	[24,33]

Table 63: Wyckoff bond: 24e@12j, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{2}, y, y\right]$	[1,-18]
2	$[-Y, -X, -X]$	$\left[\frac{1}{2}, -y, y\right]$	[2,-20]
3	$[-Y, X, X]$	$\left[\frac{1}{2}, y, -y\right]$	[3,-17]
4	$[Y, -X, X]$	$\left[\frac{1}{2}, -y, -y\right]$	[4,-19]
5	$[-X, Y, X]$	$\left[y, \frac{1}{2}, y\right]$	[5,-15]
6	$[-X, -Y, -X]$	$\left[y, \frac{1}{2}, -y\right]$	[6,-13]
7	$[X, -Y, X]$	$\left[-y, \frac{1}{2}, y\right]$	[7,-16]
8	$[X, Y, -X]$	$\left[-y, \frac{1}{2}, -y\right]$	[8,-14]
9	$[X, -X, Y]$	$\left[y, y, \frac{1}{2}\right]$	[9,-21]
10	$[-X, -X, -Y]$	$\left[-y, y, \frac{1}{2}\right]$	[10,-23]
11	$[X, X, -Y]$	$\left[y, -y, \frac{1}{2}\right]$	[11,-22]
12	$[-X, X, Y]$	$\left[-y, -y, \frac{1}{2}\right]$	[12,-24]
13	$[-Y, -X, X]$	$\left[\frac{1}{2}, -y, -y\right]$	[25,-42]
14	$[Y, X, X]$	$\left[\frac{1}{2}, y, -y\right]$	[26,-44]
15	$[Y, -X, -X]$	$\left[\frac{1}{2}, -y, y\right]$	[27,-41]
16	$[-Y, X, -X]$	$\left[\frac{1}{2}, y, y\right]$	[28,-43]
17	$[X, -Y, -X]$	$\left[-y, \frac{1}{2}, -y\right]$	[29,-39]
18	$[X, Y, X]$	$\left[-y, \frac{1}{2}, y\right]$	[30,-37]
19	$[-X, Y, -X]$	$\left[y, \frac{1}{2}, -y\right]$	[31,-40]
20	$[-X, -Y, X]$	$\left[y, \frac{1}{2}, y\right]$	[32,-38]
21	$[-X, X, -Y]$	$\left[-y, -y, \frac{1}{2}\right]$	[33,-45]
22	$[X, X, Y]$	$\left[y, -y, \frac{1}{2}\right]$	[34,-47]
23	$[-X, -X, Y]$	$\left[-y, y, \frac{1}{2}\right]$	[35,-46]
24	$[X, -X, -Y]$	$\left[y, y, \frac{1}{2}\right]$	[36,-48]

Table 64: Wyckoff bond: `24f@12j`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[\frac{1}{2},\ y,\ y\right]$	`[1,28]`
2	$[0,\ -X,\ Y]$	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2,27]`
3	$[0,\ X,\ -Y]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3,26]`
4	$[0,\ -X,\ -Y]$	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4,25]`
5	$[Y,\ 0,\ X]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[5,32]`
6	$[Y,\ 0,\ -X]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6,31]`
7	$[-Y,\ 0,\ X]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7,30]`
8	$[-Y,\ 0,\ -X]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8,29]`
9	$[X,\ Y,\ 0]$	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9,36]`
10	$[-X,\ Y,\ 0]$	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10,35]`
11	$[X,\ -Y,\ 0]$	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11,34]`
12	$[-X,\ -Y,\ 0]$	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12,33]`
13	$[X,\ 0,\ -Y]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[13,40]`
14	$[-X,\ 0,\ -Y]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[14,39]`
15	$[X,\ 0,\ Y]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[15,38]`
16	$[-X,\ 0,\ Y]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[16,37]`
17	$[0,\ Y,\ -X]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[17,44]`
18	$[0,\ Y,\ X]$	$\left[\frac{1}{2},\ y,\ y\right]$	`[18,43]`
19	$[0,\ -Y,\ -X]$	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[19,42]`
20	$[0,\ -Y,\ X]$	$\left[\frac{1}{2},\ -y,\ y\right]$	`[20,41]`
21	$[Y,\ X,\ 0]$	$\left[y,\ y,\ \frac{1}{2}\right]$	`[21,48]`
22	$[Y,\ -X,\ 0]$	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[22,47]`
23	$[-Y,\ X,\ 0]$	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[23,46]`
24	$[-Y,\ -X,\ 0]$	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[24,45]`

Table 65: Wyckoff bond: `48g@12j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ y,\ y\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ -y,\ y\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ -y,\ -y\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[y,\ y,\ \frac{1}{2}\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[-y,\ y,\ \frac{1}{2}\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[y,\ -y,\ \frac{1}{2}\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[-y,\ -y,\ \frac{1}{2}\right]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[y,\ \frac{1}{2},\ -y\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[-y,\ \frac{1}{2},\ -y\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[y,\ \frac{1}{2},\ y\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[-y,\ \frac{1}{2},\ y\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[\frac{1}{2},\ y,\ -y\right]$	`[17]`

continued ...

Table 65

No.	vector	center	mapping
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, y, y\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, -y, -y\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{1}{2}, -y, y\right]$	[20]
21	$[Z, Y, -X]$	$\left[y, y, \frac{1}{2}\right]$	[21]
22	$[Z, -Y, X]$	$\left[y, -y, \frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[-y, y, \frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[-y, -y, \frac{1}{2}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, -y, -y\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, -y\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, y\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, y\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[-y, \frac{1}{2}, -y\right]$	[29]
30	$[-Z, X, Y]$	$\left[-y, \frac{1}{2}, y\right]$	[30]
31	$[Z, X, -Y]$	$\left[y, \frac{1}{2}, -y\right]$	[31]
32	$[Z, -X, Y]$	$\left[y, \frac{1}{2}, y\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[-y, -y, \frac{1}{2}\right]$	[33]
34	$[Y, -Z, X]$	$\left[y, -y, \frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[-y, y, \frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[y, y, \frac{1}{2}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[-y, \frac{1}{2}, y\right]$	[37]
38	$[Y, X, Z]$	$\left[y, \frac{1}{2}, y\right]$	[38]
39	$[-Y, X, -Z]$	$\left[-y, \frac{1}{2}, -y\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y, \frac{1}{2}, -y\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}, -y, y\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, -y, -y\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{1}{2}, y, y\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}, y, -y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[-y, -y, \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[-y, y, \frac{1}{2}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[y, -y, \frac{1}{2}\right]$	[47]
48	$[Z, Y, X]$	$\left[y, y, \frac{1}{2}\right]$	[48]

* Wyckoff site: 24k, site symmetry: m..

Table 66: Wyckoff bond: 24a@24k, bond symmetry: m..

No.	vector	center	mapping
1	$[0, X, Y]$	$[0, y, z]$	[1,28]
2	$[0, -X, Y]$	$[0, -y, z]$	[2,27]
3	$[0, X, -Y]$	$[0, y, -z]$	[3,26]
4	$[0, -X, -Y]$	$[0, -y, -z]$	[4,25]
5	$[Y, 0, X]$	$[z, 0, y]$	[5,32]
6	$[Y, 0, -X]$	$[z, 0, -y]$	[6,31]
7	$[-Y, 0, X]$	$[-z, 0, y]$	[7,30]
8	$[-Y, 0, -X]$	$[-z, 0, -y]$	[8,29]
9	$[X, Y, 0]$	$[y, z, 0]$	[9,36]
10	$[-X, Y, 0]$	$[-y, z, 0]$	[10,35]

continued ...

Table 66

No.	vector	center	mapping
11	$[X, -Y, 0]$	$[y, -z, 0]$	[11,34]
12	$[-X, -Y, 0]$	$[-y, -z, 0]$	[12,33]
13	$[X, 0, -Y]$	$[y, 0, -z]$	[13,40]
14	$[-X, 0, -Y]$	$[-y, 0, -z]$	[14,39]
15	$[X, 0, Y]$	$[y, 0, z]$	[15,38]
16	$[-X, 0, Y]$	$[-y, 0, z]$	[16,37]
17	$[0, Y, -X]$	$[0, z, -y]$	[17,44]
18	$[0, Y, X]$	$[0, z, y]$	[18,43]
19	$[0, -Y, -X]$	$[0, -z, -y]$	[19,42]
20	$[0, -Y, X]$	$[0, -z, y]$	[20,41]
21	$[Y, X, 0]$	$[z, y, 0]$	[21,48]
22	$[Y, -X, 0]$	$[z, -y, 0]$	[22,47]
23	$[-Y, X, 0]$	$[-z, y, 0]$	[23,46]
24	$[-Y, -X, 0]$	$[-z, -y, 0]$	[24,45]

Table 67: Wyckoff bond: 24b@24k, bond symmetry: m..

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	[1,-28]
2	$[-X, 0, 0]$	$[0, -y, z]$	[2,-27]
3	$[-X, 0, 0]$	$[0, y, -z]$	[3,-26]
4	$[X, 0, 0]$	$[0, -y, -z]$	[4,-25]
5	$[0, X, 0]$	$[z, 0, y]$	[5,-32]
6	$[0, -X, 0]$	$[z, 0, -y]$	[6,-31]
7	$[0, -X, 0]$	$[-z, 0, y]$	[7,-30]
8	$[0, X, 0]$	$[-z, 0, -y]$	[8,-29]
9	$[0, 0, X]$	$[y, z, 0]$	[9,-36]
10	$[0, 0, -X]$	$[-y, z, 0]$	[10,-35]
11	$[0, 0, -X]$	$[y, -z, 0]$	[11,-34]
12	$[0, 0, X]$	$[-y, -z, 0]$	[12,-33]
13	$[0, X, 0]$	$[y, 0, -z]$	[13,-40]
14	$[0, -X, 0]$	$[-y, 0, -z]$	[14,-39]
15	$[0, -X, 0]$	$[y, 0, z]$	[15,-38]
16	$[0, X, 0]$	$[-y, 0, z]$	[16,-37]
17	$[X, 0, 0]$	$[0, z, -y]$	[17,-44]
18	$[-X, 0, 0]$	$[0, z, y]$	[18,-43]
19	$[-X, 0, 0]$	$[0, -z, -y]$	[19,-42]
20	$[X, 0, 0]$	$[0, -z, y]$	[20,-41]
21	$[0, 0, -X]$	$[z, y, 0]$	[21,-48]
22	$[0, 0, X]$	$[z, -y, 0]$	[22,-47]
23	$[0, 0, X]$	$[-z, y, 0]$	[23,-46]
24	$[0, 0, -X]$	$[-z, -y, 0]$	[24,-45]

Table 68: Wyckoff bond: `48c@24k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, z]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -z]$	`[4]`
5	$[Z, X, Y]$	$[z, 0, y]$	`[5]`
6	$[Z, -X, -Y]$	$[z, 0, -y]$	`[6]`
7	$[-Z, -X, Y]$	$[-z, 0, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, 0, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, z, 0]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, z, 0]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -z, 0]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -z, 0]$	`[12]`
13	$[Y, X, -Z]$	$[y, 0, -z]$	`[13]`
14	$[-Y, -X, -Z]$	$[-y, 0, -z]$	`[14]`
15	$[Y, -X, Z]$	$[y, 0, z]$	`[15]`
16	$[-Y, X, Z]$	$[-y, 0, z]$	`[16]`
17	$[X, Z, -Y]$	$[0, z, -y]$	`[17]`
18	$[-X, Z, Y]$	$[0, z, y]$	`[18]`
19	$[-X, -Z, -Y]$	$[0, -z, -y]$	`[19]`
20	$[X, -Z, Y]$	$[0, -z, y]$	`[20]`
21	$[Z, Y, -X]$	$[z, y, 0]$	`[21]`
22	$[Z, -Y, X]$	$[z, -y, 0]$	`[22]`
23	$[-Z, Y, X]$	$[-z, y, 0]$	`[23]`
24	$[-Z, -Y, -X]$	$[-z, -y, 0]$	`[24]`
25	$[-X, -Y, -Z]$	$[0, -y, -z]$	`[25]`
26	$[X, Y, -Z]$	$[0, y, -z]$	`[26]`
27	$[X, -Y, Z]$	$[0, -y, z]$	`[27]`
28	$[-X, Y, Z]$	$[0, y, z]$	`[28]`
29	$[-Z, -X, -Y]$	$[-z, 0, -y]$	`[29]`
30	$[-Z, X, Y]$	$[-z, 0, y]$	`[30]`
31	$[Z, X, -Y]$	$[z, 0, -y]$	`[31]`
32	$[Z, -X, Y]$	$[z, 0, y]$	`[32]`
33	$[-Y, -Z, -X]$	$[-y, -z, 0]$	`[33]`
34	$[Y, -Z, X]$	$[y, -z, 0]$	`[34]`
35	$[-Y, Z, X]$	$[-y, z, 0]$	`[35]`
36	$[Y, Z, -X]$	$[y, z, 0]$	`[36]`
37	$[-Y, -X, Z]$	$[-y, 0, z]$	`[37]`
38	$[Y, X, Z]$	$[y, 0, z]$	`[38]`
39	$[-Y, X, -Z]$	$[-y, 0, -z]$	`[39]`
40	$[Y, -X, -Z]$	$[y, 0, -z]$	`[40]`
41	$[-X, -Z, Y]$	$[0, -z, y]$	`[41]`
42	$[X, -Z, -Y]$	$[0, -z, -y]$	`[42]`
43	$[X, Z, Y]$	$[0, z, y]$	`[43]`
44	$[-X, Z, -Y]$	$[0, z, -y]$	`[44]`
45	$[-Z, -Y, X]$	$[-z, -y, 0]$	`[45]`
46	$[-Z, Y, -X]$	$[-z, y, 0]$	`[46]`

continued ...

Table 68

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$[z, -y, 0]$	`[47]`
48	$[Z, Y, X]$	$[z, y, 0]$	`[48]`

* Wyckoff site: `24l`, site symmetry: `m..`

Table 69: Wyckoff bond: `24a@24l`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[\frac{1}{2}, y, z\right]$	`[1,28]`
2	$[0, -X, Y]$	$\left[\frac{1}{2}, -y, z\right]$	`[2,27]`
3	$[0, X, -Y]$	$\left[\frac{1}{2}, y, -z\right]$	`[3,26]`
4	$[0, -X, -Y]$	$\left[\frac{1}{2}, -y, -z\right]$	`[4,25]`
5	$[Y, 0, X]$	$\left[z, \frac{1}{2}, y\right]$	`[5,32]`
6	$[Y, 0, -X]$	$\left[z, \frac{1}{2}, -y\right]$	`[6,31]`
7	$[-Y, 0, X]$	$\left[-z, \frac{1}{2}, y\right]$	`[7,30]`
8	$[-Y, 0, -X]$	$\left[-z, \frac{1}{2}, -y\right]$	`[8,29]`
9	$[X, Y, 0]$	$\left[y, z, \frac{1}{2}\right]$	`[9,36]`
10	$[-X, Y, 0]$	$\left[-y, z, \frac{1}{2}\right]$	`[10,35]`
11	$[X, -Y, 0]$	$\left[y, -z, \frac{1}{2}\right]$	`[11,34]`
12	$[-X, -Y, 0]$	$\left[-y, -z, \frac{1}{2}\right]$	`[12,33]`
13	$[X, 0, -Y]$	$\left[y, \frac{1}{2}, -z\right]$	`[13,40]`
14	$[-X, 0, -Y]$	$\left[-y, \frac{1}{2}, -z\right]$	`[14,39]`
15	$[X, 0, Y]$	$\left[y, \frac{1}{2}, z\right]$	`[15,38]`
16	$[-X, 0, Y]$	$\left[-y, \frac{1}{2}, z\right]$	`[16,37]`
17	$[0, Y, -X]$	$\left[\frac{1}{2}, z, -y\right]$	`[17,44]`
18	$[0, Y, X]$	$\left[\frac{1}{2}, z, y\right]$	`[18,43]`
19	$[0, -Y, -X]$	$\left[\frac{1}{2}, -z, -y\right]$	`[19,42]`
20	$[0, -Y, X]$	$\left[\frac{1}{2}, -z, y\right]$	`[20,41]`
21	$[Y, X, 0]$	$\left[z, y, \frac{1}{2}\right]$	`[21,48]`
22	$[Y, -X, 0]$	$\left[z, -y, \frac{1}{2}\right]$	`[22,47]`
23	$[-Y, X, 0]$	$\left[-z, y, \frac{1}{2}\right]$	`[23,46]`
24	$[-Y, -X, 0]$	$\left[-z, -y, \frac{1}{2}\right]$	`[24,45]`

Table 70: Wyckoff bond: `24b@24l`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{2}, y, z\right]$	`[1,-28]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, -y, z\right]$	`[2,-27]`
3	$[-X, 0, 0]$	$\left[\frac{1}{2}, y, -z\right]$	`[3,-26]`
4	$[X, 0, 0]$	$\left[\frac{1}{2}, -y, -z\right]$	`[4,-25]`
5	$[0, X, 0]$	$\left[z, \frac{1}{2}, y\right]$	`[5,-32]`
6	$[0, -X, 0]$	$\left[z, \frac{1}{2}, -y\right]$	`[6,-31]`
7	$[0, -X, 0]$	$\left[-z, \frac{1}{2}, y\right]$	`[7,-30]`
8	$[0, X, 0]$	$\left[-z, \frac{1}{2}, -y\right]$	`[8,-29]`
9	$[0, 0, X]$	$\left[y, z, \frac{1}{2}\right]$	`[9,-36]`
10	$[0, 0, -X]$	$\left[-y, z, \frac{1}{2}\right]$	`[10,-35]`

continued ...

Table 70

No.	vector	center	mapping
11	$[0,\ 0,\ -X]$	$\left[y,\ -z,\ \frac{1}{2}\right]$	`[11,-34]`
12	$[0,\ 0,\ X]$	$\left[-y,\ -z,\ \frac{1}{2}\right]$	`[12,-33]`
13	$[0,\ X,\ 0]$	$\left[y,\ \frac{1}{2},\ -z\right]$	`[13,-40]`
14	$[0,\ -X,\ 0]$	$\left[-y,\ \frac{1}{2},\ -z\right]$	`[14,-39]`
15	$[0,\ -X,\ 0]$	$\left[y,\ \frac{1}{2},\ z\right]$	`[15,-38]`
16	$[0,\ X,\ 0]$	$\left[-y,\ \frac{1}{2},\ z\right]$	`[16,-37]`
17	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ z,\ -y\right]$	`[17,-44]`
18	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ z,\ y\right]$	`[18,-43]`
19	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ -z,\ -y\right]$	`[19,-42]`
20	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ -z,\ y\right]$	`[20,-41]`
21	$[0,\ 0,\ -X]$	$\left[z,\ y,\ \frac{1}{2}\right]$	`[21,-48]`
22	$[0,\ 0,\ X]$	$\left[z,\ -y,\ \frac{1}{2}\right]$	`[22,-47]`
23	$[0,\ 0,\ X]$	$\left[-z,\ y,\ \frac{1}{2}\right]$	`[23,-46]`
24	$[0,\ 0,\ -X]$	$\left[-z,\ -y,\ \frac{1}{2}\right]$	`[24,-45]`

Table 71: Wyckoff bond: `48c@241`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ y,\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ -y,\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ y,\ -z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ -y,\ -z\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[z,\ \frac{1}{2},\ y\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[z,\ \frac{1}{2},\ -y\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[-z,\ \frac{1}{2},\ y\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[-z,\ \frac{1}{2},\ -y\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[y,\ z,\ \frac{1}{2}\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[-y,\ z,\ \frac{1}{2}\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[y,\ -z,\ \frac{1}{2}\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[-y,\ -z,\ \frac{1}{2}\right]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[y,\ \frac{1}{2},\ -z\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[-y,\ \frac{1}{2},\ -z\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[y,\ \frac{1}{2},\ z\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[-y,\ \frac{1}{2},\ z\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[\frac{1}{2},\ z,\ -y\right]$	`[17]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{1}{2},\ z,\ y\right]$	`[18]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{1}{2},\ -z,\ -y\right]$	`[19]`
20	$[X,\ -Z,\ Y]$	$\left[\frac{1}{2},\ -z,\ y\right]$	`[20]`
21	$[Z,\ Y,\ -X]$	$\left[z,\ y,\ \frac{1}{2}\right]$	`[21]`
22	$[Z,\ -Y,\ X]$	$\left[z,\ -y,\ \frac{1}{2}\right]$	`[22]`
23	$[-Z,\ Y,\ X]$	$\left[-z,\ y,\ \frac{1}{2}\right]$	`[23]`
24	$[-Z,\ -Y,\ -X]$	$\left[-z,\ -y,\ \frac{1}{2}\right]$	`[24]`
25	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{2},\ -y,\ -z\right]$	`[25]`
26	$[X,\ Y,\ -Z]$	$\left[\frac{1}{2},\ y,\ -z\right]$	`[26]`
27	$[X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ -y,\ z\right]$	`[27]`

continued ...

Table 71

No.	vector	center	mapping
28	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, z\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[-z, \frac{1}{2}, -y\right]$	[29]
30	$[-Z, X, Y]$	$\left[-z, \frac{1}{2}, y\right]$	[30]
31	$[Z, X, -Y]$	$\left[z, \frac{1}{2}, -y\right]$	[31]
32	$[Z, -X, Y]$	$\left[z, \frac{1}{2}, y\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[-y, -z, \frac{1}{2}\right]$	[33]
34	$[Y, -Z, X]$	$\left[y, -z, \frac{1}{2}\right]$	[34]
35	$[-Y, Z, X]$	$\left[-y, z, \frac{1}{2}\right]$	[35]
36	$[Y, Z, -X]$	$\left[y, z, \frac{1}{2}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[-y, \frac{1}{2}, z\right]$	[37]
38	$[Y, X, Z]$	$\left[y, \frac{1}{2}, z\right]$	[38]
39	$[-Y, X, -Z]$	$\left[-y, \frac{1}{2}, -z\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y, \frac{1}{2}, -z\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}, -z, y\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{1}{2}, -z, -y\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{1}{2}, z, y\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}, z, -y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[-z, -y, \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[-z, y, \frac{1}{2}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[z, -y, \frac{1}{2}\right]$	[47]
48	$[Z, Y, X]$	$\left[z, y, \frac{1}{2}\right]$	[48]

* Wyckoff site: 24m, site symmetry: ..m

Table 72: Wyckoff bond: 24a@24m, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, X, Y]$	$[x, x, z]$	[1,38]
2	$[-X, -X, Y]$	$[-x, -x, z]$	[2,37]
3	$[-X, X, -Y]$	$[-x, x, -z]$	[3,39]
4	$[X, -X, -Y]$	$[x, -x, -z]$	[4,40]
5	$[Y, X, X]$	$[z, x, x]$	[5,48]
6	$[Y, -X, -X]$	$[z, -x, -x]$	[6,47]
7	$[-Y, -X, X]$	$[-z, -x, x]$	[7,45]
8	$[-Y, X, -X]$	$[-z, x, -x]$	[8,46]
9	$[X, Y, X]$	$[x, z, x]$	[9,43]
10	$[-X, Y, -X]$	$[-x, z, -x]$	[10,44]
11	$[X, -Y, -X]$	$[x, -z, -x]$	[11,42]
12	$[-X, -Y, X]$	$[-x, -z, x]$	[12,41]
13	$[X, X, -Y]$	$[x, x, -z]$	[13,26]
14	$[-X, -X, -Y]$	$[-x, -x, -z]$	[14,25]
15	$[X, -X, Y]$	$[x, -x, z]$	[15,27]
16	$[-X, X, Y]$	$[-x, x, z]$	[16,28]
17	$[X, Y, -X]$	$[x, z, -x]$	[17,36]
18	$[-X, Y, X]$	$[-x, z, x]$	[18,35]
19	$[-X, -Y, -X]$	$[-x, -z, -x]$	[19,33]
20	$[X, -Y, X]$	$[x, -z, x]$	[20,34]

continued ...

Table 72

No.	vector	center	mapping
21	$[Y, X, -X]$	$[z, x, -x]$	[21,31]
22	$[Y, -X, X]$	$[z, -x, x]$	[22,32]
23	$[-Y, X, X]$	$[-z, x, x]$	[23,30]
24	$[-Y, -X, -X]$	$[-z, -x, -x]$	[24,29]

Table 73: Wyckoff bond: 24b@24m, bond symmetry: ..m

No.	vector	center	mapping
1	$[X, -X, 0]$	$[x, x, z]$	[1,-38]
2	$[-X, X, 0]$	$[-x, -x, z]$	[2,-37]
3	$[-X, -X, 0]$	$[-x, x, -z]$	[3,-39]
4	$[X, X, 0]$	$[x, -x, -z]$	[4,-40]
5	$[0, X, -X]$	$[z, x, x]$	[5,-48]
6	$[0, -X, X]$	$[z, -x, -x]$	[6,-47]
7	$[0, -X, -X]$	$[-z, -x, x]$	[7,-45]
8	$[0, X, X]$	$[-z, x, -x]$	[8,-46]
9	$[-X, 0, X]$	$[x, z, x]$	[9,-43]
10	$[X, 0, -X]$	$[-x, z, -x]$	[10,-44]
11	$[-X, 0, -X]$	$[x, -z, -x]$	[11,-42]
12	$[X, 0, X]$	$[-x, -z, x]$	[12,-41]
13	$[-X, X, 0]$	$[x, x, -z]$	[13,-26]
14	$[X, -X, 0]$	$[-x, -x, -z]$	[14,-25]
15	$[-X, -X, 0]$	$[x, -x, z]$	[15,-27]
16	$[X, X, 0]$	$[-x, x, z]$	[16,-28]
17	$[X, 0, X]$	$[x, z, -x]$	[17,-36]
18	$[-X, 0, -X]$	$[-x, z, x]$	[18,-35]
19	$[-X, 0, X]$	$[-x, -z, -x]$	[19,-33]
20	$[X, 0, -X]$	$[x, -z, x]$	[20,-34]
21	$[0, -X, -X]$	$[z, x, -x]$	[21,-31]
22	$[0, X, X]$	$[z, -x, x]$	[22,-32]
23	$[0, -X, X]$	$[-z, x, x]$	[23,-30]
24	$[0, X, -X]$	$[-z, -x, -x]$	[24,-29]

Table 74: Wyckoff bond: 48c@24m, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -x, z]$	[2]
3	$[-X, Y, -Z]$	$[-x, x, -z]$	[3]
4	$[X, -Y, -Z]$	$[x, -x, -z]$	[4]
5	$[Z, X, Y]$	$[z, x, x]$	[5]
6	$[Z, -X, -Y]$	$[z, -x, -x]$	[6]
7	$[-Z, -X, Y]$	$[-z, -x, x]$	[7]
8	$[-Z, X, -Y]$	$[-z, x, -x]$	[8]

continued ...

Table 74

No.	vector	center	mapping
9	$[Y, Z, X]$	$[x, z, x]$	[9]
10	$[-Y, Z, -X]$	$[-x, z, -x]$	[10]
11	$[Y, -Z, -X]$	$[x, -z, -x]$	[11]
12	$[-Y, -Z, X]$	$[-x, -z, x]$	[12]
13	$[Y, X, -Z]$	$[x, x, -z]$	[13]
14	$[-Y, -X, -Z]$	$[-x, -x, -z]$	[14]
15	$[Y, -X, Z]$	$[x, -x, z]$	[15]
16	$[-Y, X, Z]$	$[-x, x, z]$	[16]
17	$[X, Z, -Y]$	$[x, z, -x]$	[17]
18	$[-X, Z, Y]$	$[-x, z, x]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -z, -x]$	[19]
20	$[X, -Z, Y]$	$[x, -z, x]$	[20]
21	$[Z, Y, -X]$	$[z, x, -x]$	[21]
22	$[Z, -Y, X]$	$[z, -x, x]$	[22]
23	$[-Z, Y, X]$	$[-z, x, x]$	[23]
24	$[-Z, -Y, -X]$	$[-z, -x, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -x, -z]$	[25]
26	$[X, Y, -Z]$	$[x, x, -z]$	[26]
27	$[X, -Y, Z]$	$[x, -x, z]$	[27]
28	$[-X, Y, Z]$	$[-x, x, z]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -x]$	[29]
30	$[-Z, X, Y]$	$[-z, x, x]$	[30]
31	$[Z, X, -Y]$	$[z, x, -x]$	[31]
32	$[Z, -X, Y]$	$[z, -x, x]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -z, -x]$	[33]
34	$[Y, -Z, X]$	$[x, -z, x]$	[34]
35	$[-Y, Z, X]$	$[-x, z, x]$	[35]
36	$[Y, Z, -X]$	$[x, z, -x]$	[36]
37	$[-Y, -X, Z]$	$[-x, -x, z]$	[37]
38	$[Y, X, Z]$	$[x, x, z]$	[38]
39	$[-Y, X, -Z]$	$[-x, x, -z]$	[39]
40	$[Y, -X, -Z]$	$[x, -x, -z]$	[40]
41	$[-X, -Z, Y]$	$[-x, -z, x]$	[41]
42	$[X, -Z, -Y]$	$[x, -z, -x]$	[42]
43	$[X, Z, Y]$	$[x, z, x]$	[43]
44	$[-X, Z, -Y]$	$[-x, z, -x]$	[44]
45	$[-Z, -Y, X]$	$[-z, -x, x]$	[45]
46	$[-Z, Y, -X]$	$[-z, x, -x]$	[46]
47	$[Z, -Y, -X]$	$[z, -x, -x]$	[47]
48	$[Z, Y, X]$	$[z, x, x]$	[48]

* Wyckoff site: 48n, site symmetry: 1

Table 75: Wyckoff bond: `48a@48n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -y, z]$	[2]
3	$[-X, Y, -Z]$	$[-x, y, -z]$	[3]
4	$[X, -Y, -Z]$	$[x, -y, -z]$	[4]
5	$[Z, X, Y]$	$[z, x, y]$	[5]
6	$[Z, -X, -Y]$	$[z, -x, -y]$	[6]
7	$[-Z, -X, Y]$	$[-z, -x, y]$	[7]
8	$[-Z, X, -Y]$	$[-z, x, -y]$	[8]
9	$[Y, Z, X]$	$[y, z, x]$	[9]
10	$[-Y, Z, -X]$	$[-y, z, -x]$	[10]
11	$[Y, -Z, -X]$	$[y, -z, -x]$	[11]
12	$[-Y, -Z, X]$	$[-y, -z, x]$	[12]
13	$[Y, X, -Z]$	$[y, x, -z]$	[13]
14	$[-Y, -X, -Z]$	$[-y, -x, -z]$	[14]
15	$[Y, -X, Z]$	$[y, -x, z]$	[15]
16	$[-Y, X, Z]$	$[-y, x, z]$	[16]
17	$[X, Z, -Y]$	$[x, z, -y]$	[17]
18	$[-X, Z, Y]$	$[-x, z, y]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -z, -y]$	[19]
20	$[X, -Z, Y]$	$[x, -z, y]$	[20]
21	$[Z, Y, -X]$	$[z, y, -x]$	[21]
22	$[Z, -Y, X]$	$[z, -y, x]$	[22]
23	$[-Z, Y, X]$	$[-z, y, x]$	[23]
24	$[-Z, -Y, -X]$	$[-z, -y, -x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[25]
26	$[X, Y, -Z]$	$[x, y, -z]$	[26]
27	$[X, -Y, Z]$	$[x, -y, z]$	[27]
28	$[-X, Y, Z]$	$[-x, y, z]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -y]$	[29]
30	$[-Z, X, Y]$	$[-z, x, y]$	[30]
31	$[Z, X, -Y]$	$[z, x, -y]$	[31]
32	$[Z, -X, Y]$	$[z, -x, y]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -z, -x]$	[33]
34	$[Y, -Z, X]$	$[y, -z, x]$	[34]
35	$[-Y, Z, X]$	$[-y, z, x]$	[35]
36	$[Y, Z, -X]$	$[y, z, -x]$	[36]
37	$[-Y, -X, Z]$	$[-y, -x, z]$	[37]
38	$[Y, X, Z]$	$[y, x, z]$	[38]
39	$[-Y, X, -Z]$	$[-y, x, -z]$	[39]
40	$[Y, -X, -Z]$	$[y, -x, -z]$	[40]
41	$[-X, -Z, Y]$	$[-x, -z, y]$	[41]
42	$[X, -Z, -Y]$	$[x, -z, -y]$	[42]
43	$[X, Z, Y]$	$[x, z, y]$	[43]
44	$[-X, Z, -Y]$	$[-x, z, -y]$	[44]
45	$[-Z, -Y, X]$	$[-z, -y, x]$	[45]
46	$[-Z, Y, -X]$	$[-z, y, -x]$	[46]

continued ...

Table 75

No.	vector	center	mapping
47	$[Z, -Y, -X]$	$[z, -y, -x]$	[47]
48	$[Z, Y, X]$	$[z, y, x]$	[48]